



MAT 208: Vector Spaces

- Prof. Dr. Pitikhate Sooraksa

Week 1:

- Introduction to the class
- Motivation for studying this class
- Knowledge from the class



About the instructor:

Adjunct faculty, CMKL,

Board Member of AIEI,

Professor @ Department of Robotics and AI, KMITL



AUTONOMOUS INTELLIGENT VEHICLE (AIV)

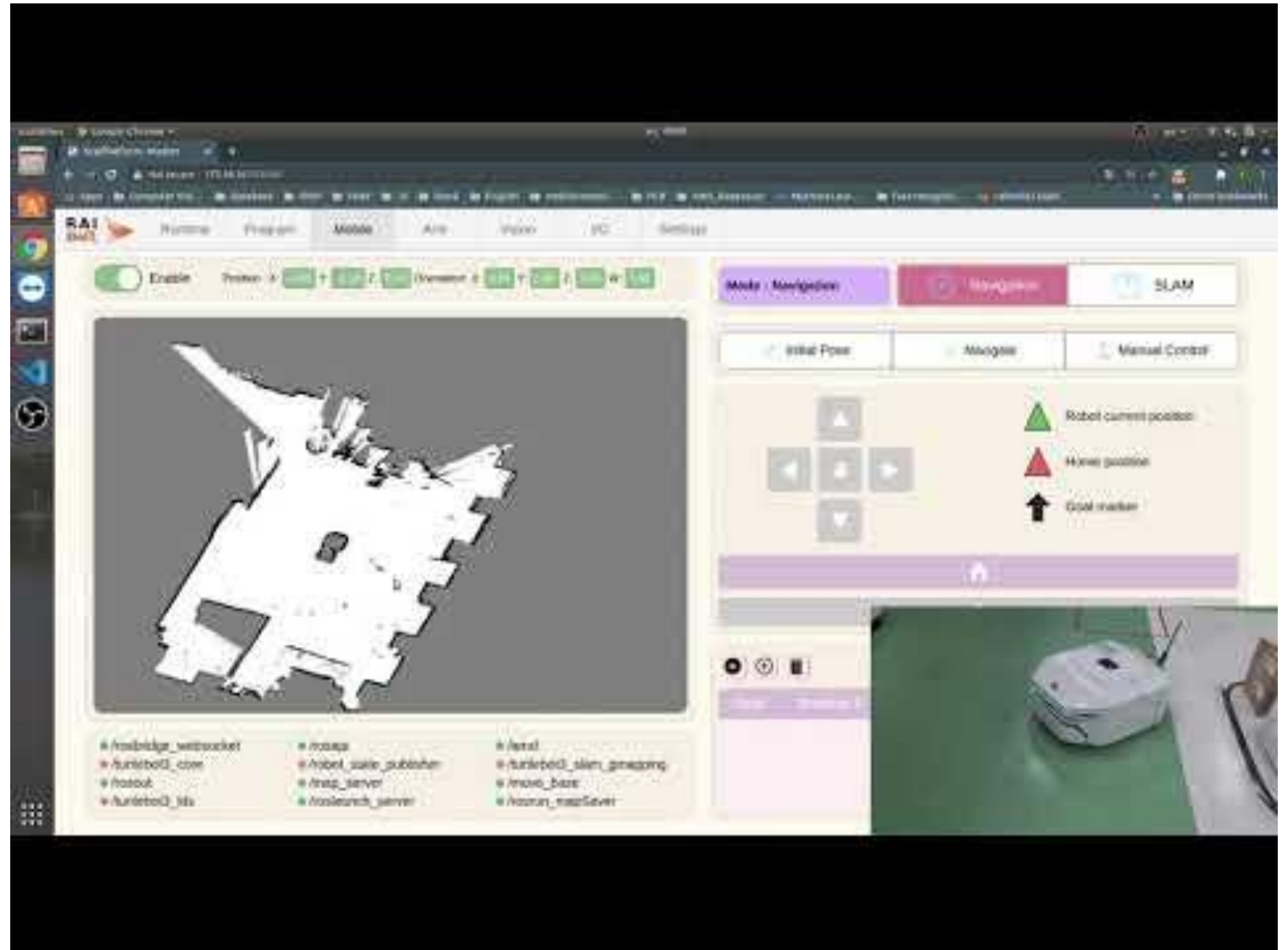
“ Industrial MOBILE
ROBOT in HDD clean room
passed SOP39 safety
standard”



AIV robot by KMITL-Seagate
Dr. Somvat Kaitwanidvilai

MOBILE ROBOT PLATFORM

“ User friendly usage for
education and industrial
MOBILE ROBOT ”





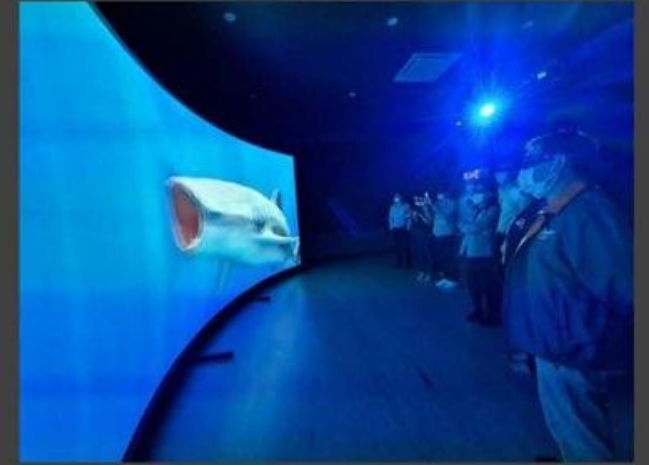
**MAC HSA LOADING
SYSTEM FOR HSA
METROLOGY TOOLS**



Machine Vision System for Copper Clad
Laminate Surface Defect Detection



MACHINE VISION SYSTEM FOR COPPER CLAD LAMINATE SURFACE DEFECT DETECTION



KIDC Interactive
Digital Center
KMITL, Bangkok



<https://www.facebook.com/KMITL.IDC/>

This course and the class:



Course Description:

A vector space is a set whose elements, vectors, satisfy specific properties.

Matrices can be formally introduced as representations of linear transformations acting on vector spaces.

Thus, the study of vector spaces provides the fundamentals for understanding more advanced topics in mathematics, physics, and computer science, where matrices play a major role.



Pillars: Mathematics

Version: V5

Version Status: Active Prime

Domain: Mathematics for AI

Start Date: 2026-08-17

End Date: 2026-09-14

Instructor: Pitikhate Sooraksa

Credits: 4-- REQUIRED

Prerequisites: -- None –

Distribution Areas: Mathematics



Competencies:

MAT-208:00010

- Calculating basic operations for vectors and matrices
- Solve basic vector and matrix equations, including additions/subtraction, dot product, matrix multiplication (by block), transpose, etc.

Topic: Vectors and Matrices

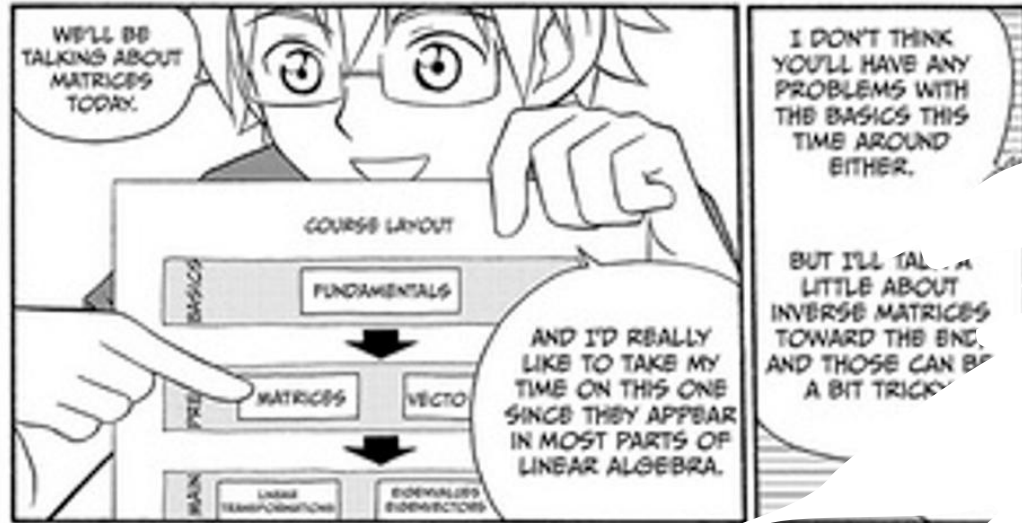


Competencies:

MAT-208:00020

- Determining basic matrices of linear transformations
- Given a basic linear transformation (e.g., rotation), design a matrix equivalent to the transformation and vice versa

Topic: Linear Transformation



WHAT IS A MATRIX?

A MATRIX IS A COLLECTION OF NUMBERS ARRANGED IN m ROWS AND n COLUMNS, BOUNDED BY PARENTHESES, LIKE THIS.

	COLUMN 1	COLUMN 2	...	COLUMN n
ROW 1	a_{11}	a_{12}	$\dots$	a_{1n}
ROW 2	a_{21}	a_{22}	$\dots$	a_{2n}
...	$\vdots$	$\vdots$	$\ddots$	$\vdots$
ROW m	a_{m1}	a_{m2}	$\dots$	a_{mn}

THESE ARE CALLED SUB

How to earn a Legend title

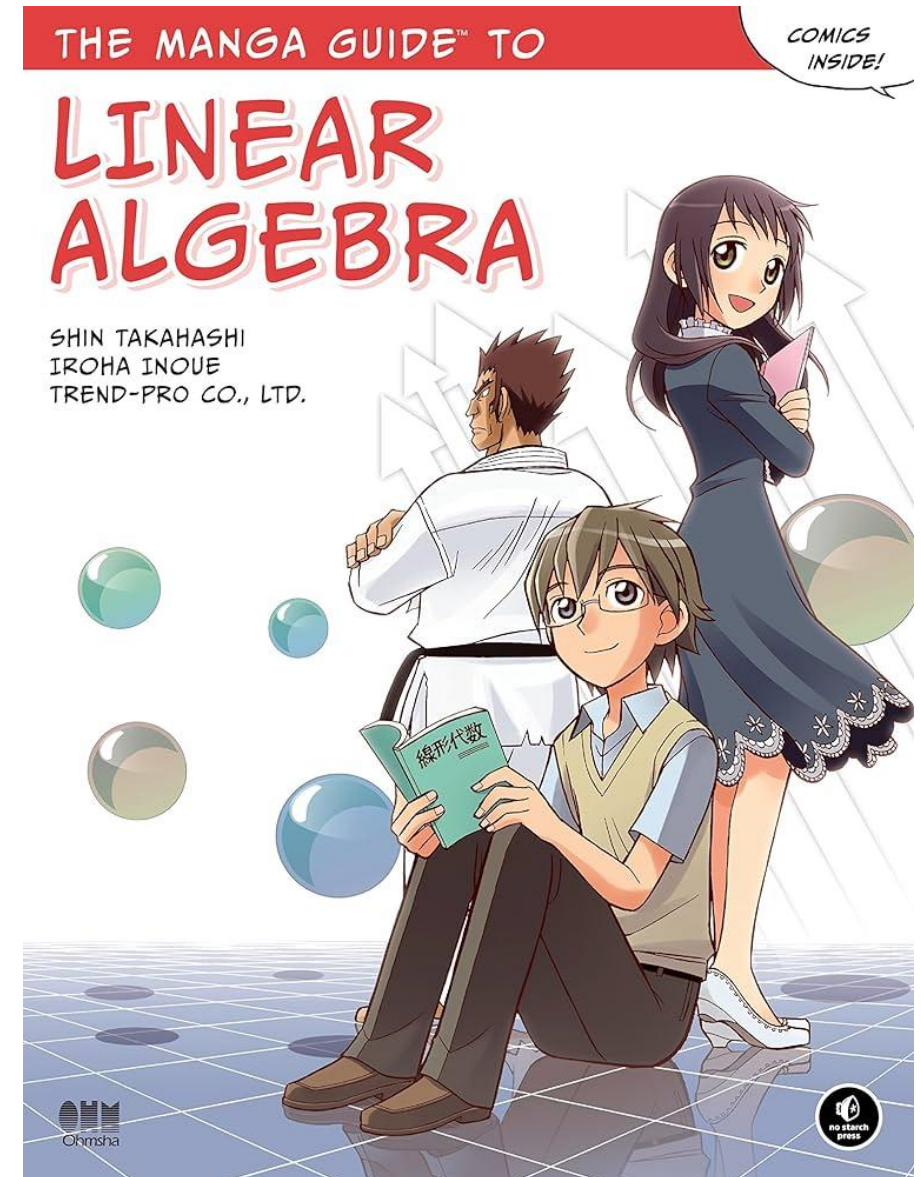
- Group study

<https://amberstudent.com/blog/post/benefits-of-group-study>

How to earn a Legend title

- Attend this class and do homework assignment

-
<https://www.mathworks.com/help/symbolic/rotation-matrix-and-transformation-matrix.html>



How to earn a Legend title

- Learn from excellent sources

- [LectureNotes.pdf \(cmu.edu\)](#)

["Linear Algebra and Applications: An Inquiry-Based Approach" by Feryal Alayont and Steven Schlicker \(gvsu.edu\)](#)

- [30. Linear Transformations and Their Matrices \(youtube.com\)](#)

[https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/](#)

Useful Resources:

- Matrices and Linear Transformation @CMU
<https://www.math.cmu.edu/~ldietric/21-241/>
- Open Courseware <https://ocw.mit.edu/courses/18-06sc-linear-algebra-fall-2011/pages/positive-definite-matrices-and-applications/linear-transformations-and-their-matrices/>
- Vector Space <https://medium.com/@cynewylf/understanding-vector-spaces-a-journey-into-linear-algebra-542816b3859b>
- Explore Vector Space <https://medium.com/@ebimsv/mastering-linear-algebra-part-5-exploring-vector-spaces-and-subspaces-9d2323df18cd>
- Matrix Laboratory <https://www.alooba.com/skills/programming-languages/programming/matrix-laboratory/>
- MATLAB <https://www.mathworks.com/products/matlab-online.html>
- Gemini, ChatGPT, Copilot, Claude, DeepSeek or any other

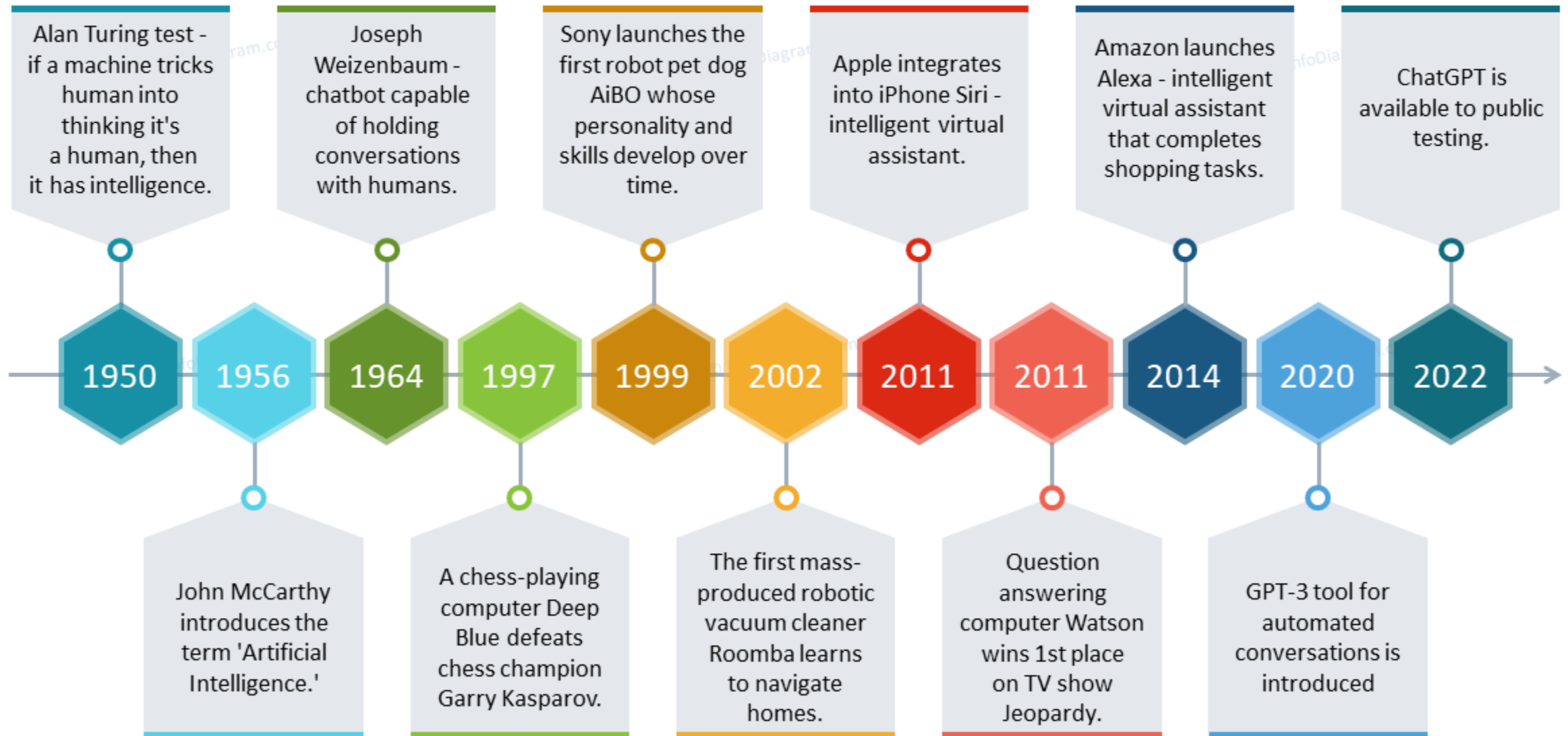
Schedule for the Class

- Week 1: (Aug 17, 2026) Introduction to MAT 208 (Vector Space)
- Week 2: (Aug 24, 2026) Concept and Basic Operations
- Week 3: (Aug 31, 2026) Vectors and Linear Transformation
- Week 4: (Sept 7, 2026) Vectors and Linear Transformation (Continue)
- Week 5: (Sept 14, 2026) Mini-project Presentation*

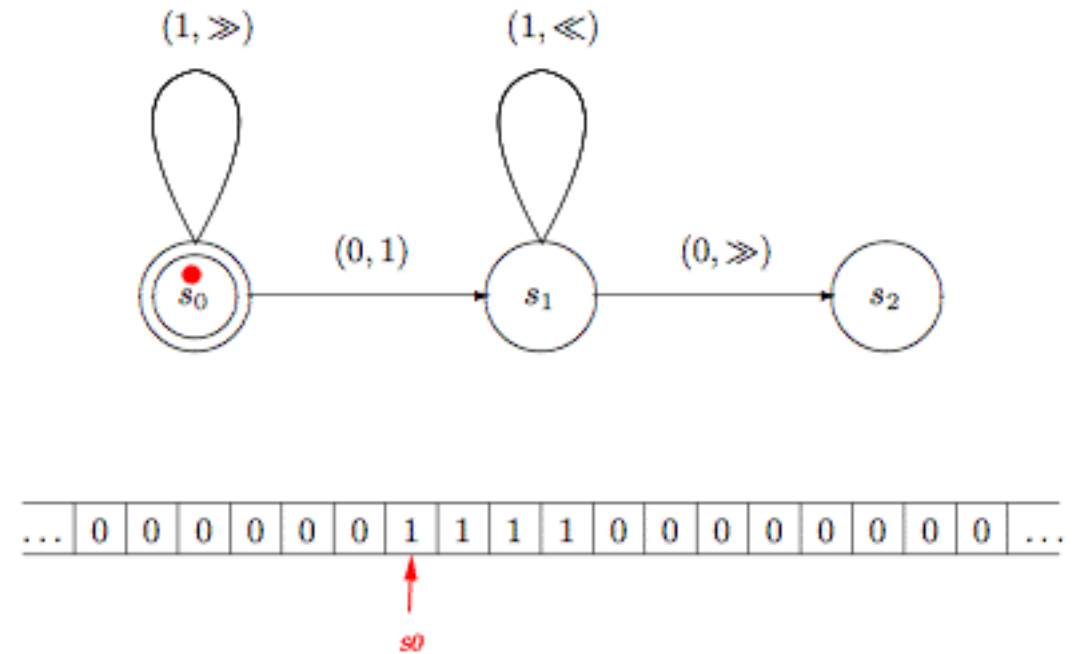
*The class has been rescheduled because the professor is attending a ceremony to receive the royal decoration of **Knight Grand Cordon (Special Class) of the Most Exalted Order of the White Elephant** during the regular lecture hour. Therefore, the session will be moved online this evening (to be announced).

Motivation related to AI:

Artificial Intelligence Development History Timeline



1948 Turing Machineent in embedded AI



The standard Turing Machine

Definition 9.1

A turing machine M is defined by

$$M = (Q, \Sigma, \Gamma, q_0, \square, F),$$

where

Q is the set of internal states,

Σ is the input alphabet,

Γ is a finite set of symbols called the ~~tape alphabet~~,

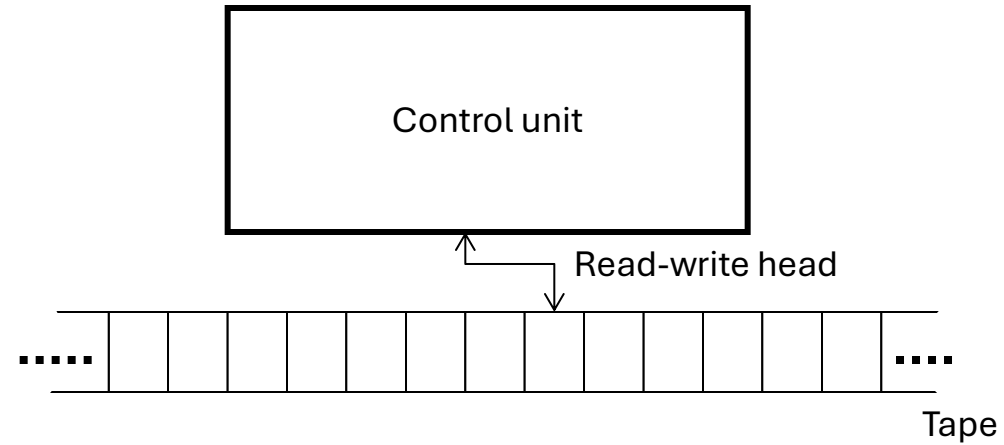
δ is the transition function,

$\square \in \Gamma$ is a special symbol called the ~~blank~~,

$q_0 \in Q$ is the initial state,

$F \subseteq Q$ is the set of final states.

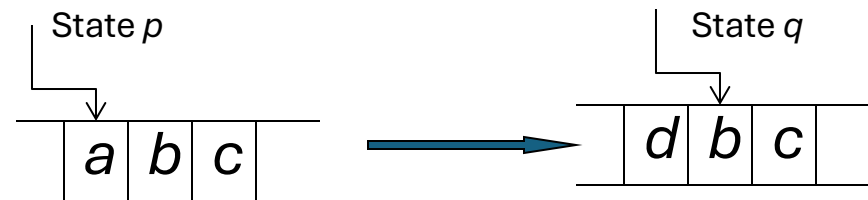
Halt state: A Turing machine will halt whenever it enters a final state or reaches a configuration for which δ is not defined.



$$\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$

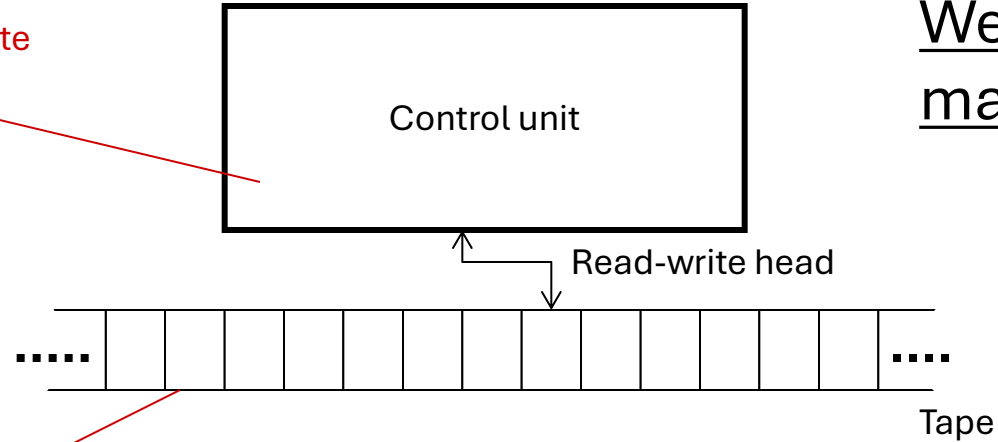
Example 9.1 The following figure shows the situation before and after the move caused by the transition .

$$\delta(p, a) = (q, d, R)$$



$$\delta(p, a) = (q, d, R)$$

Processing unit with a finite memory.



We can think of a Turing machine as a computer.

A secondary storage of unlimited capacity.

Example: Consider the Turing machine defined by

$$Q = \{q_0, q_1\},$$

$$\Sigma = \{a, b\},$$

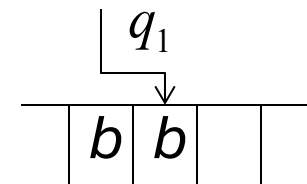
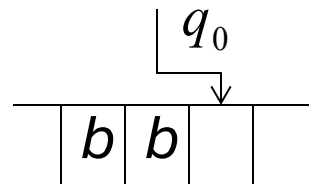
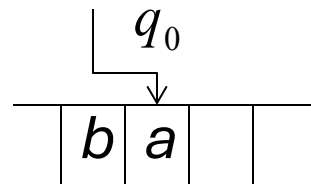
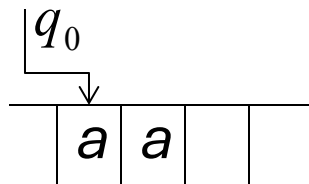
$$\Gamma = \{a, b, \square\},$$

$$F = \{q_1\}$$

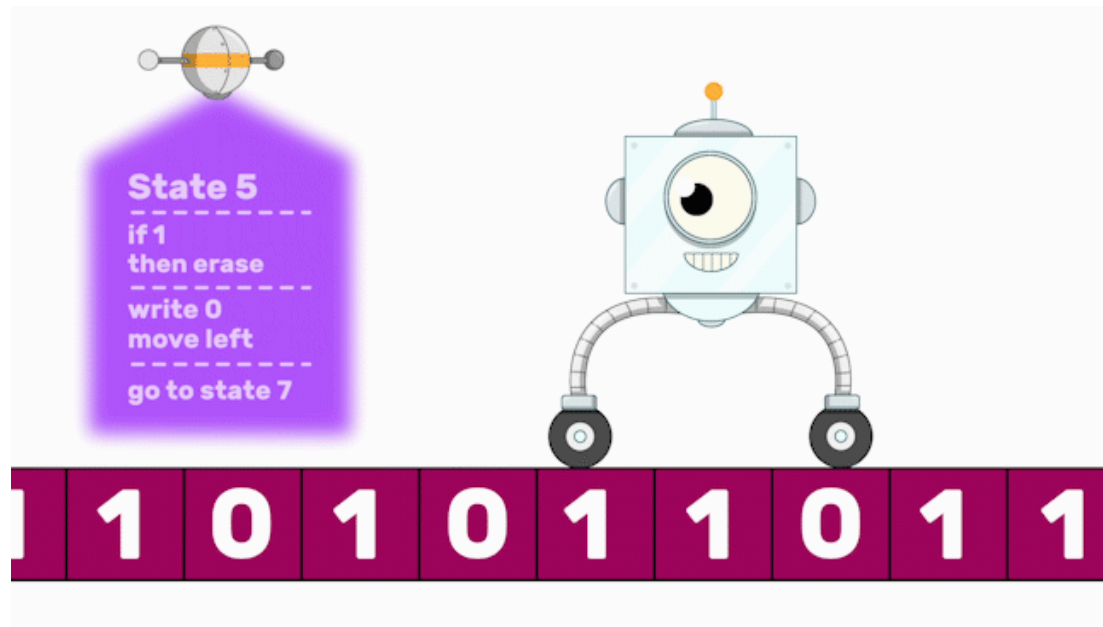
$$\delta(q_0, a) = (q_0, b, R),$$

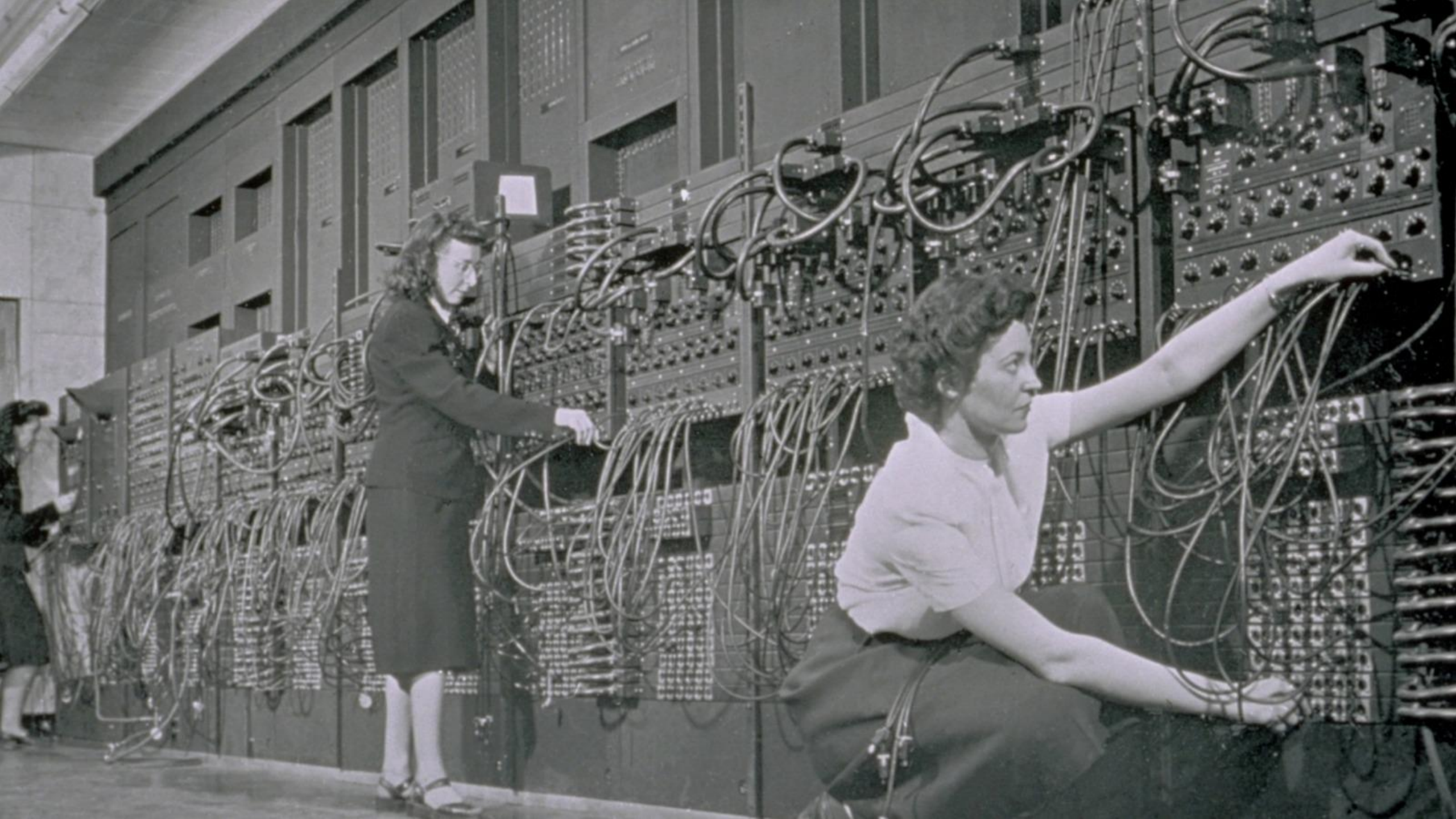
$$\delta(q_0, b) = (q_0, b, R),$$

$$\delta(q_0, \square) = (q_1, \square, L),$$



Turing Machine Concept



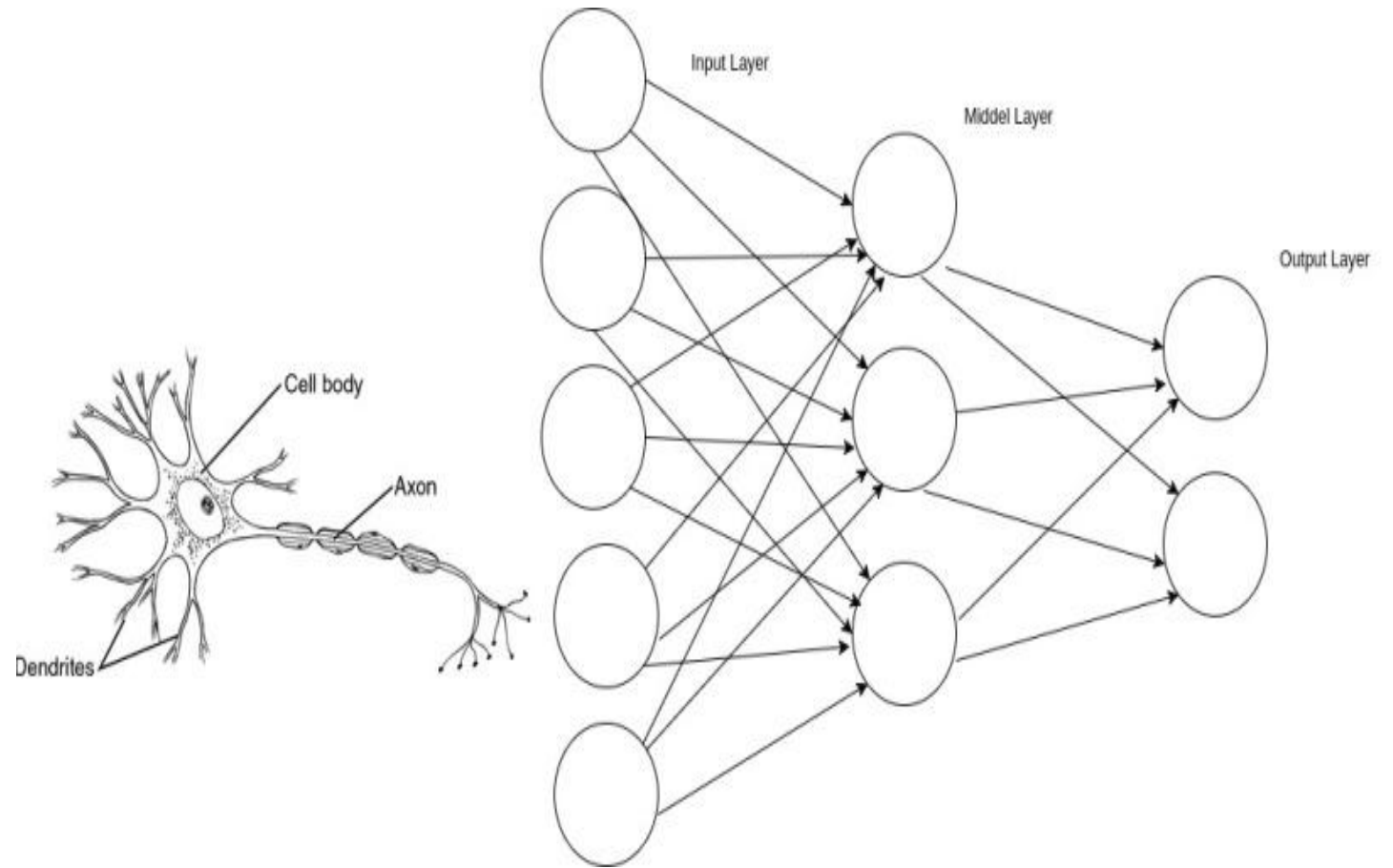


Vectors

"For decades, we treated data like a library—something to be stored and retrieved. **AI has turned that library into a living brain.**

It's **reading every book simultaneously, writing new ones, and discovering connections** we never knew existed.

You're not building shelves; **you're building the neural pathways of that brain.**"



2024 Nobel Prize

2024 Nobel Prize in physics
awarded to John J. Hopfield,
Geoffrey E. Hinton for
discoveries that 'enable
machine learning with artificial
neural networks'



John J.
Hopfield



Geoffrey E.
Hinton

October 8, 2024 Source: Princeton.edu, Toronto.edu



Recreating the brain

How an artificial neural network emulates the brain

In the brain, the neural network is made of actual nerve cells, which communicate through synapses. When we learn any new thing, some neural links get stronger and others weaker.



In an artificial neural network, interconnected nodes are trained to act like the brain's neurons, so that some of the connections get stronger while others get weaker.



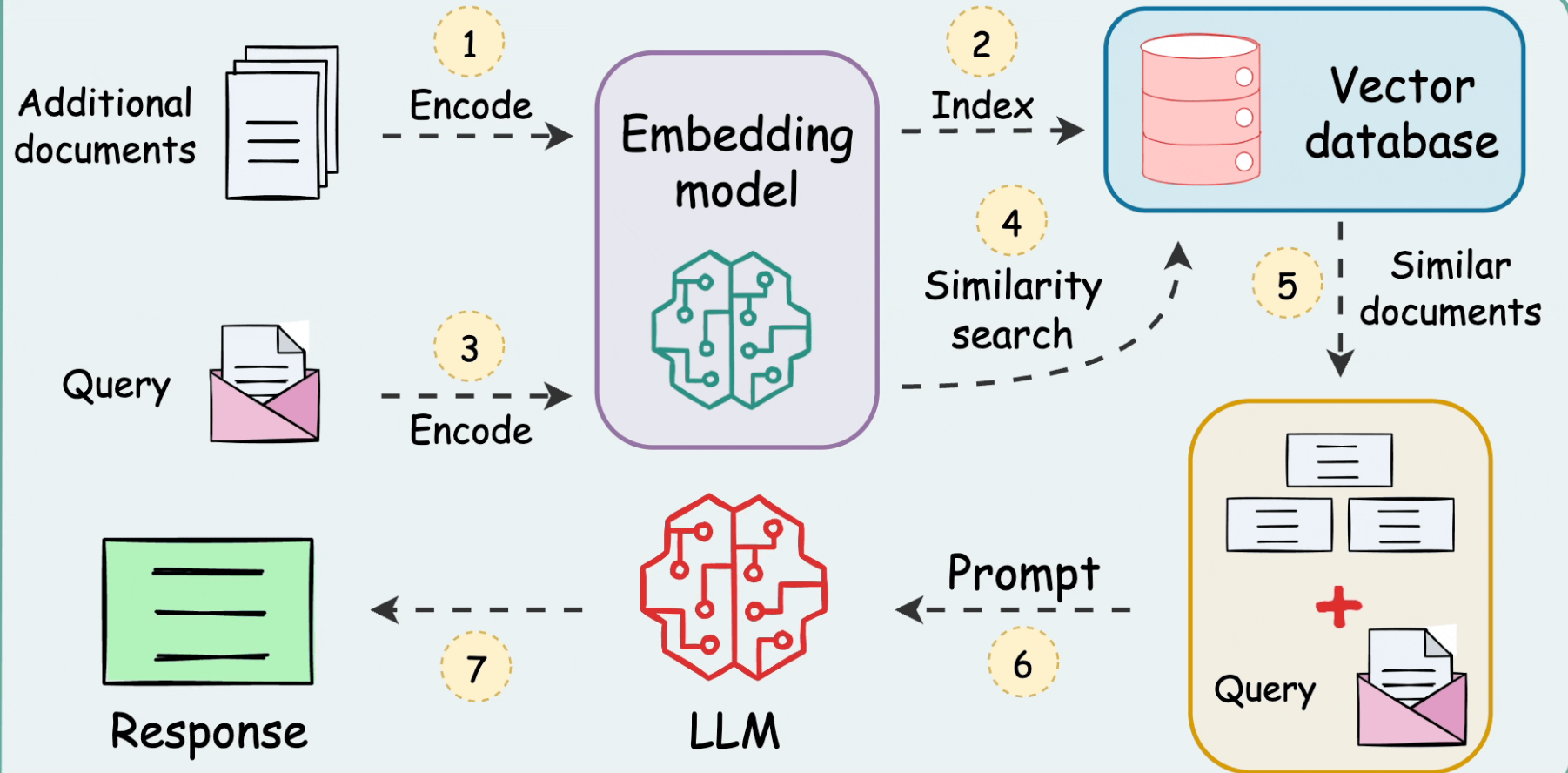
SOURCE: THE ROYAL SWEDISH ACADEMY OF SCIENCES

THE WINNERS

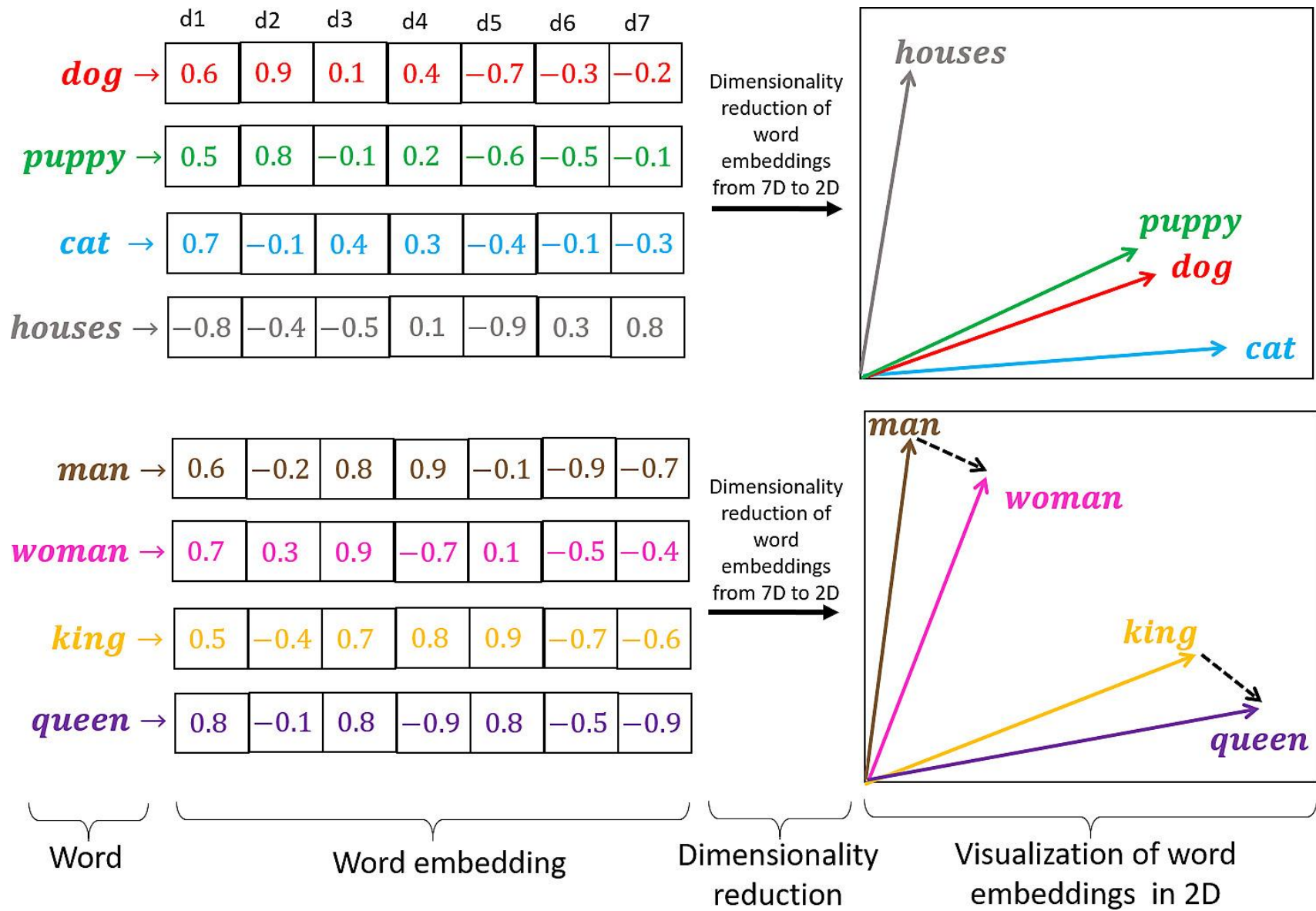
 John Hopfield, with Princeton University, created a structure that can store and reconstruct info

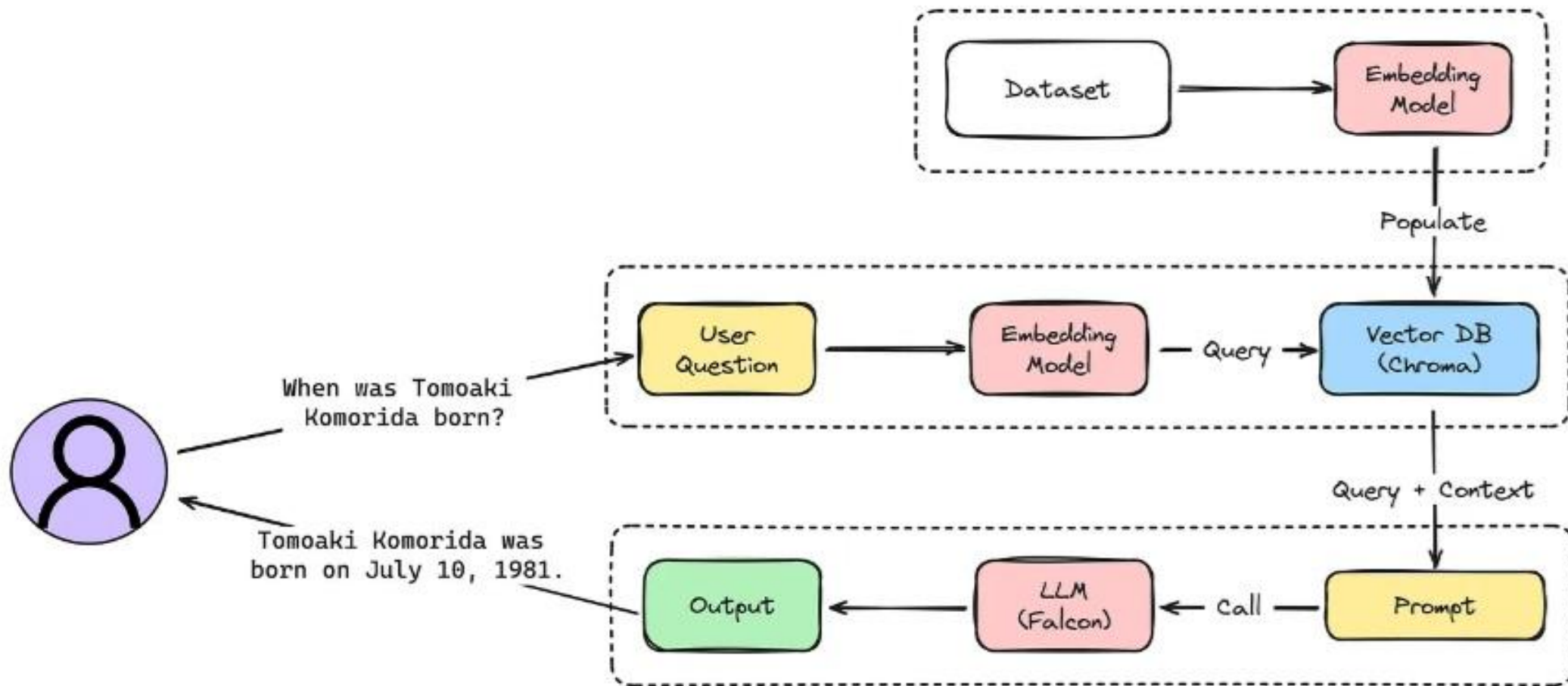
 Geoffrey Hinton, with the University of Toronto, trained a machine to generate new data based on data fed into it

RAG



The world of Large Language Models (LLMs) and their synergy with Vector Databases





Attention Is All You Need

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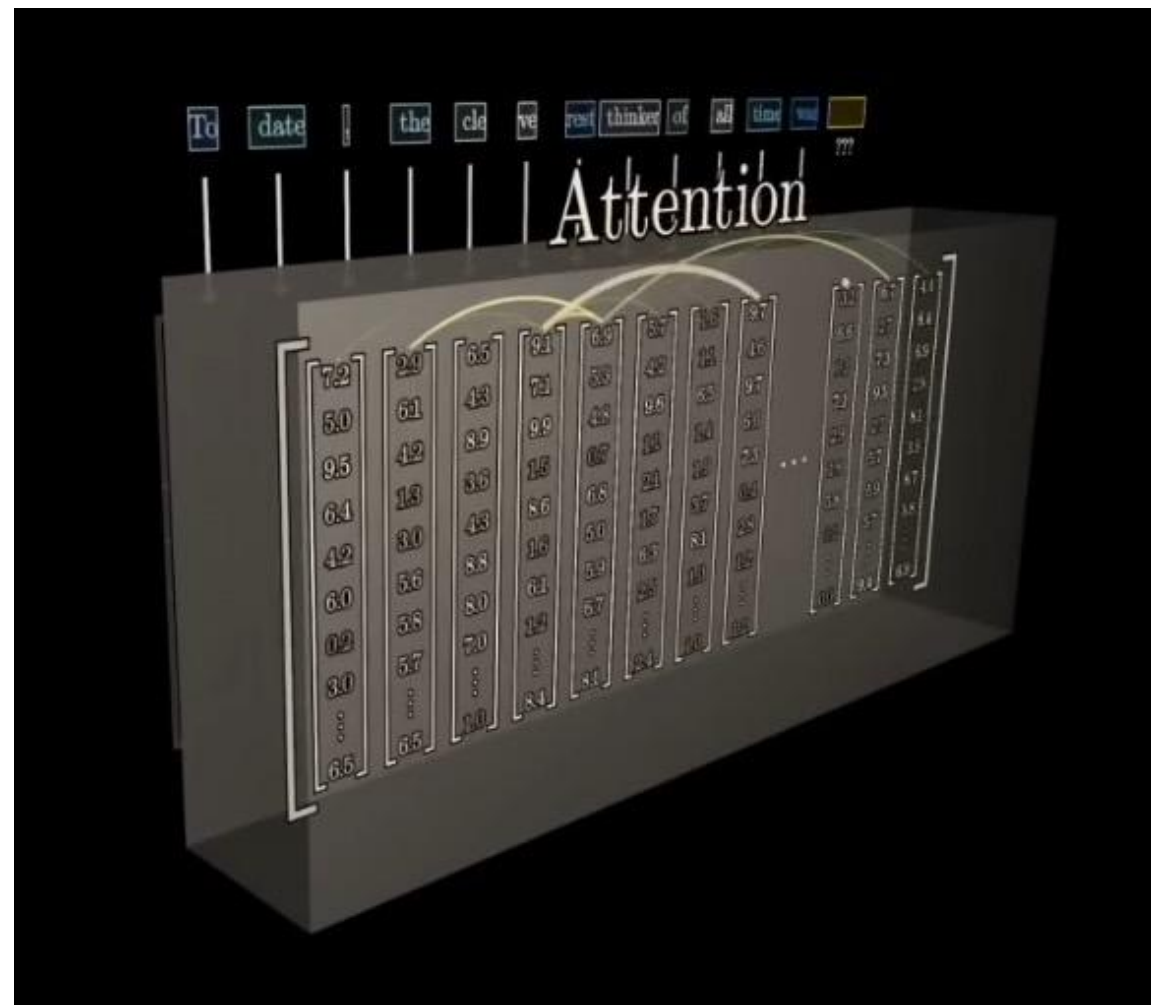
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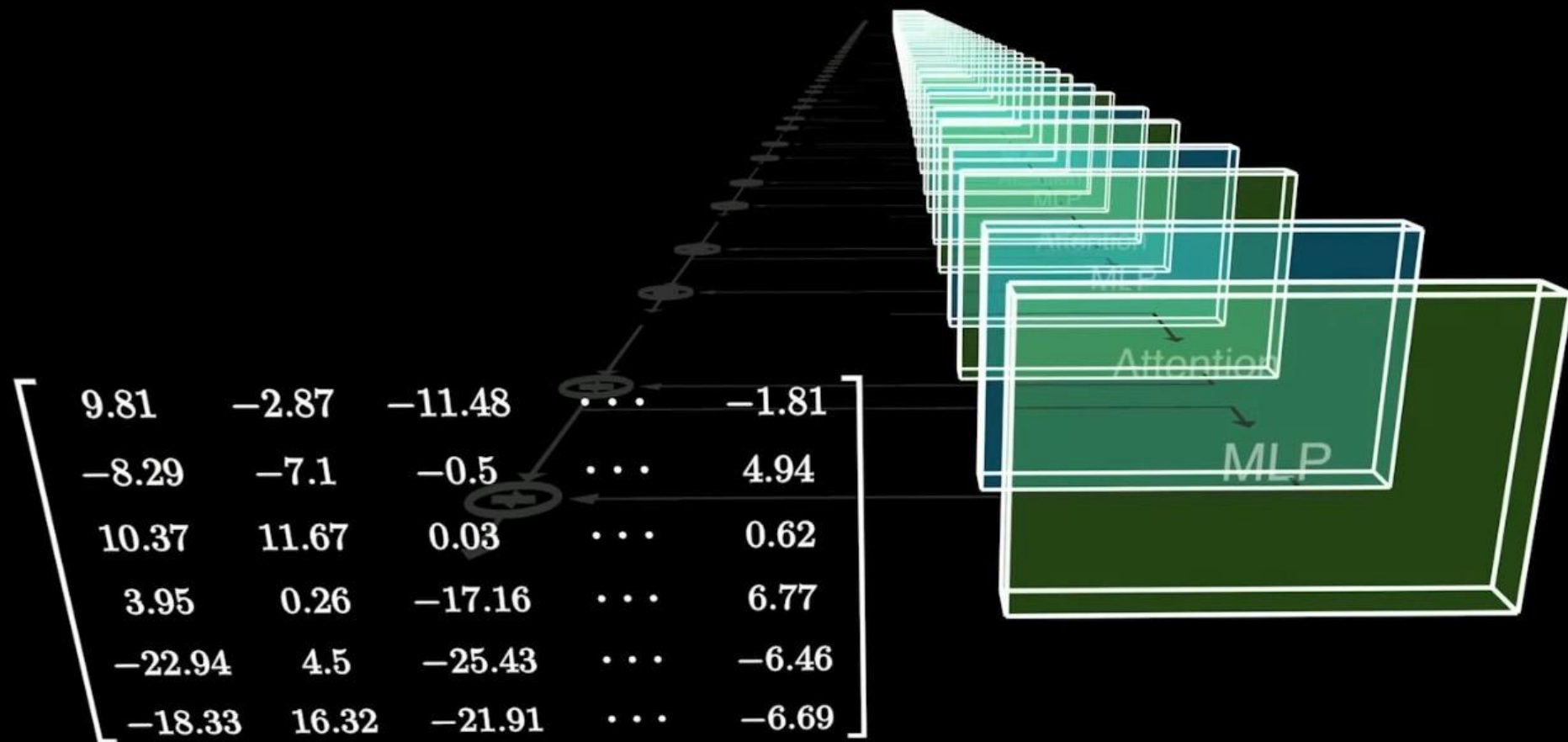
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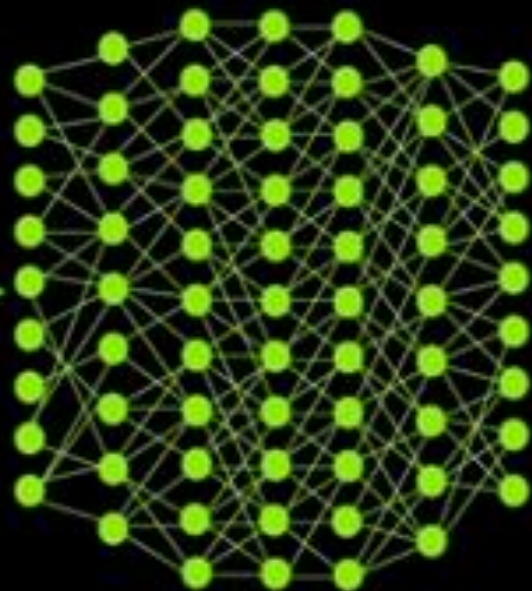
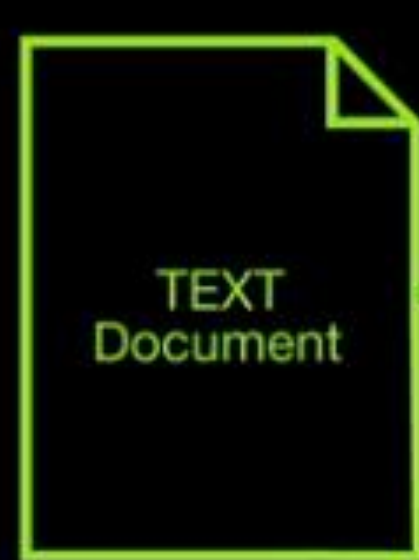
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Abstract

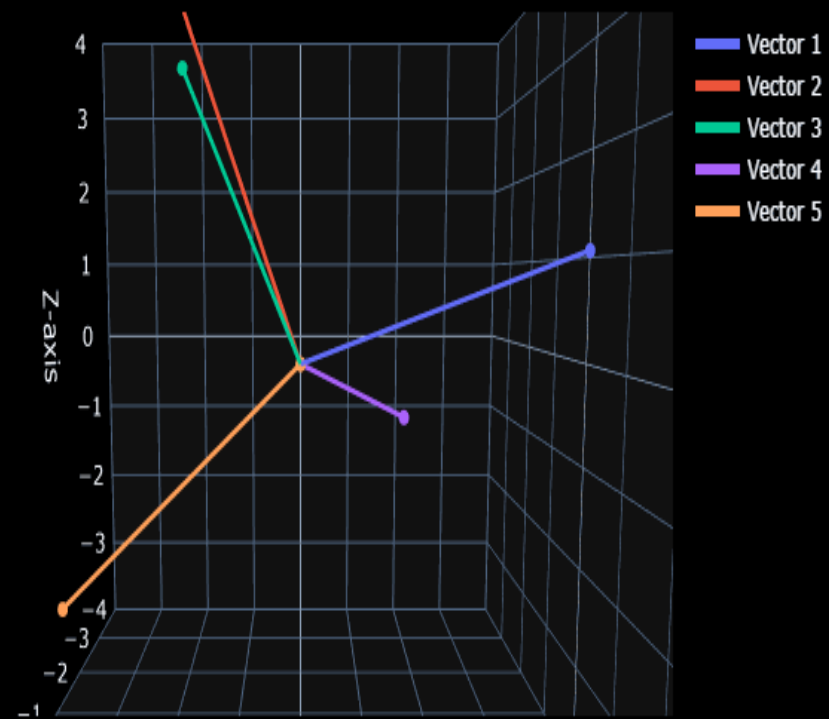
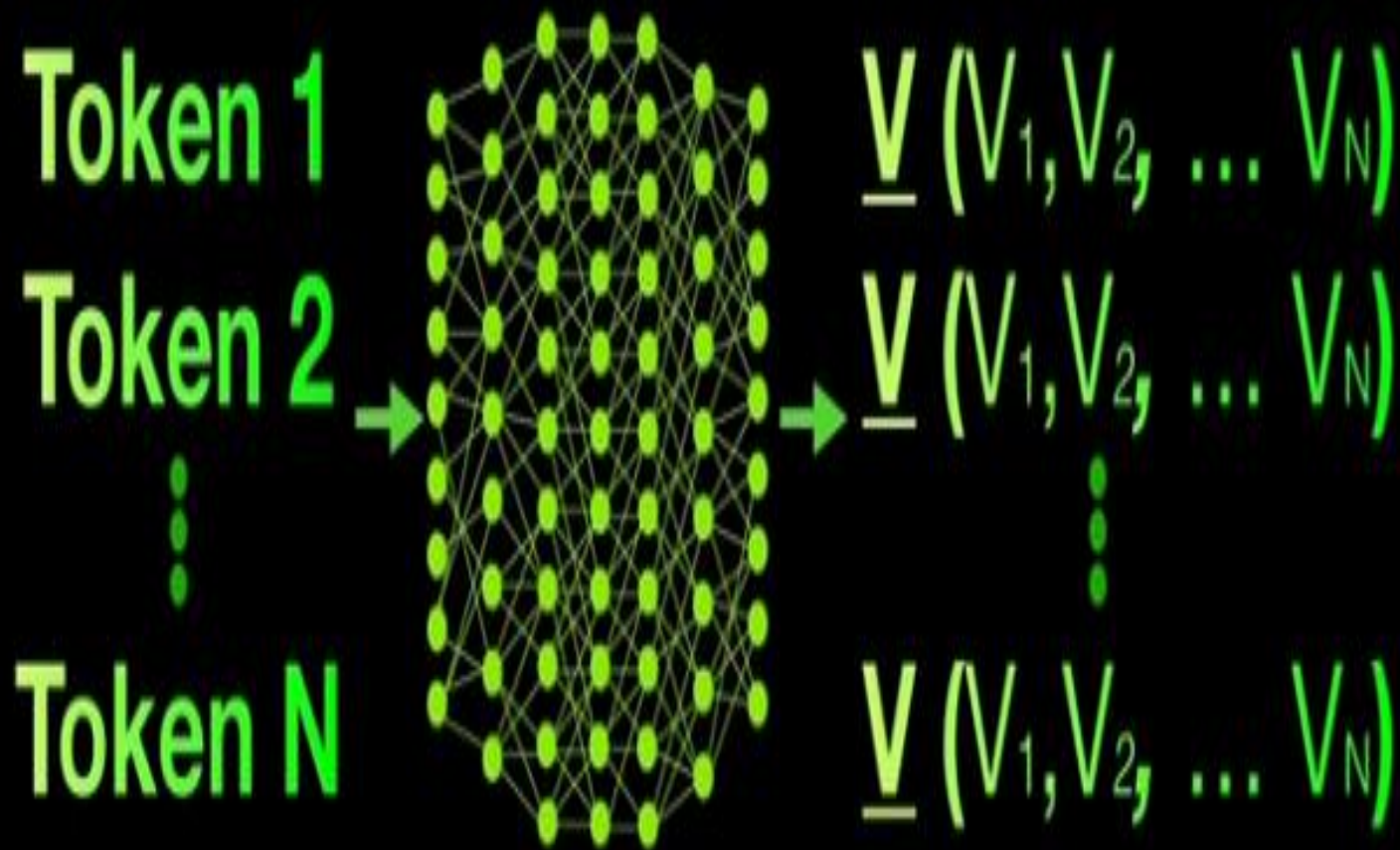
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.





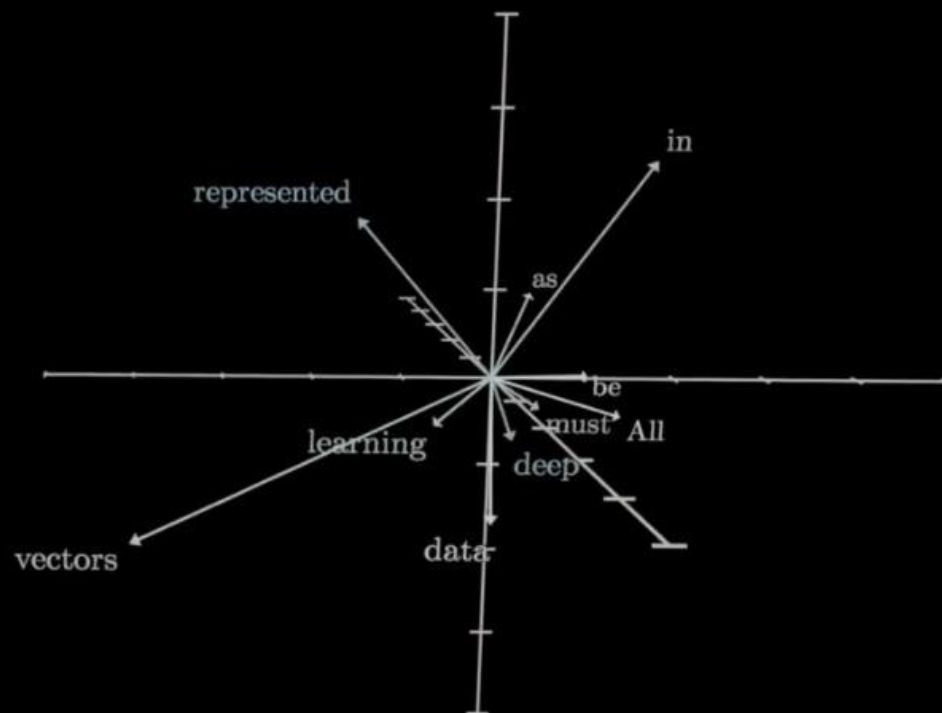


$$\begin{aligned} & \underline{\mathbf{V}} (V_1, V_2, \dots V_N) \\ & \underline{\mathbf{V}} (V_1, V_2, \dots V_N) \\ & \vdots \\ & \underline{\mathbf{V}} (V_1, V_2, \dots V_N) \end{aligned}$$



Words $\xrightarrow{\text{"Embedding"}}$ Vectors

All
data
in
deep
learning
must
be
represented
as
vectors



What is a Vector



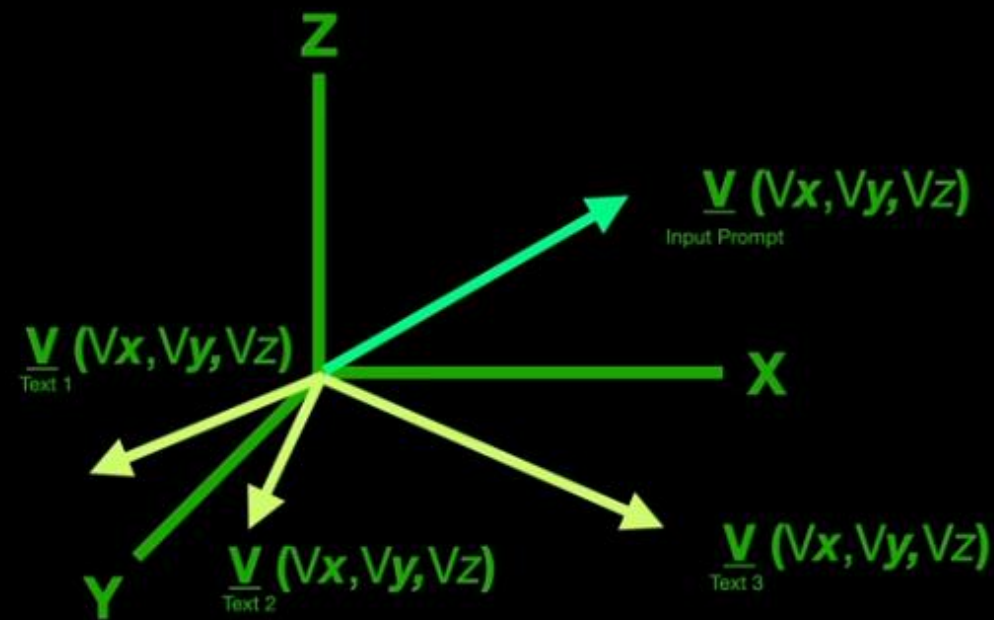
2-D $\underline{\mathbf{V}} (V_1, V_2)$

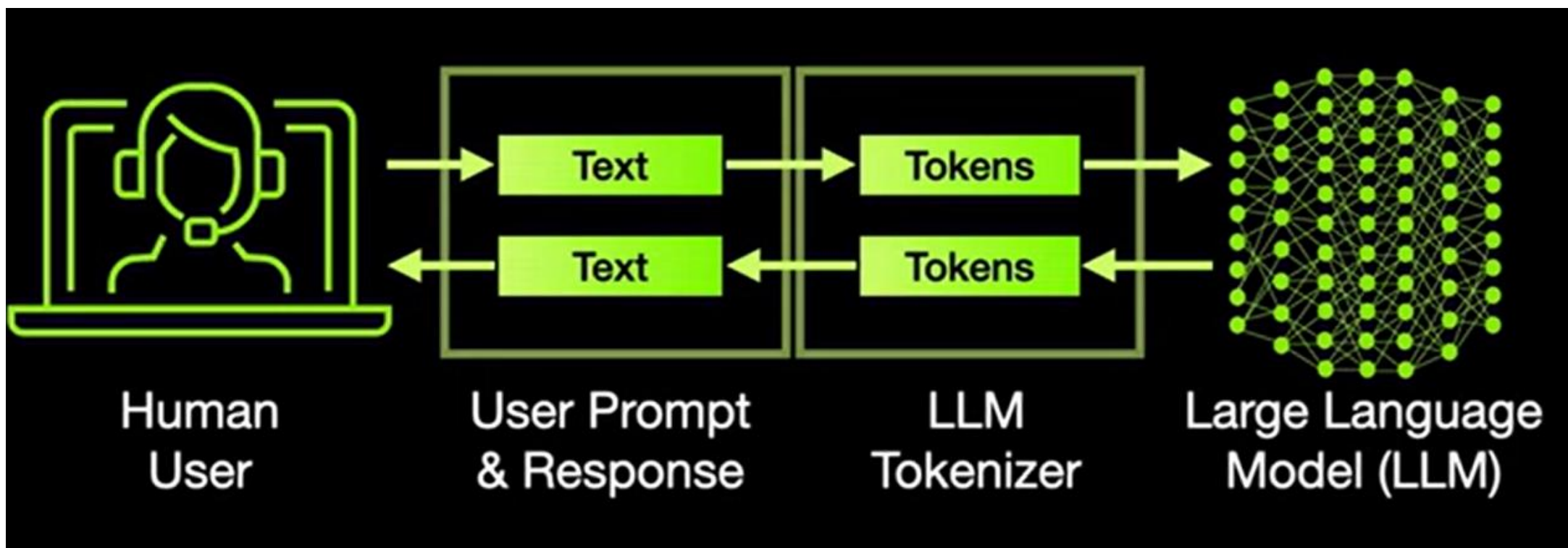
3-D $\underline{\mathbf{V}} (V_1, V_2, V_3)$

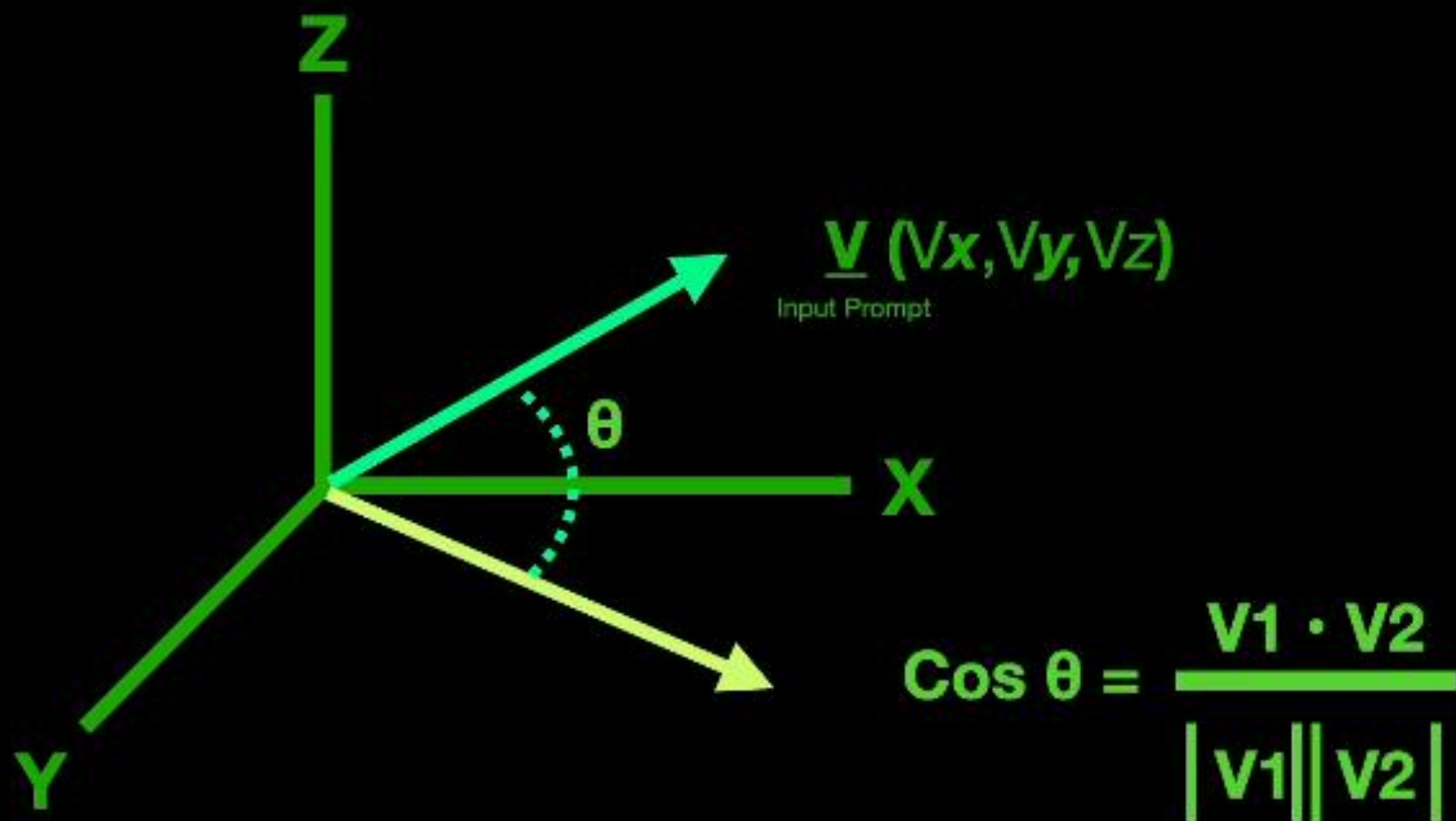
4-D $\underline{\mathbf{V}} (V_1, V_2, V_3, V_4)$

N-D $\underline{\mathbf{V}} (V_1, V_2, \dots V_{N-1}, V_N)$

LLM Similarity Searches

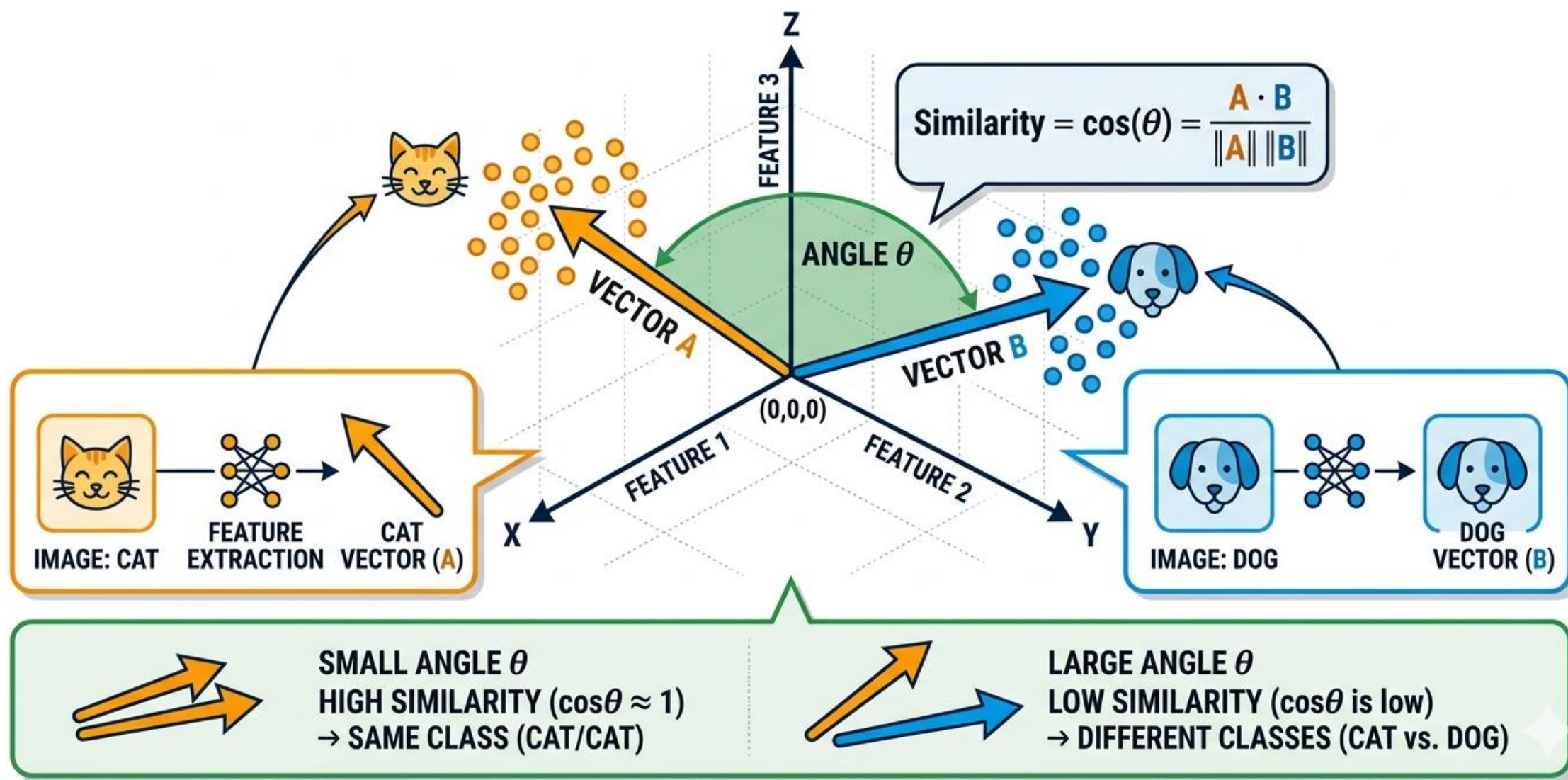




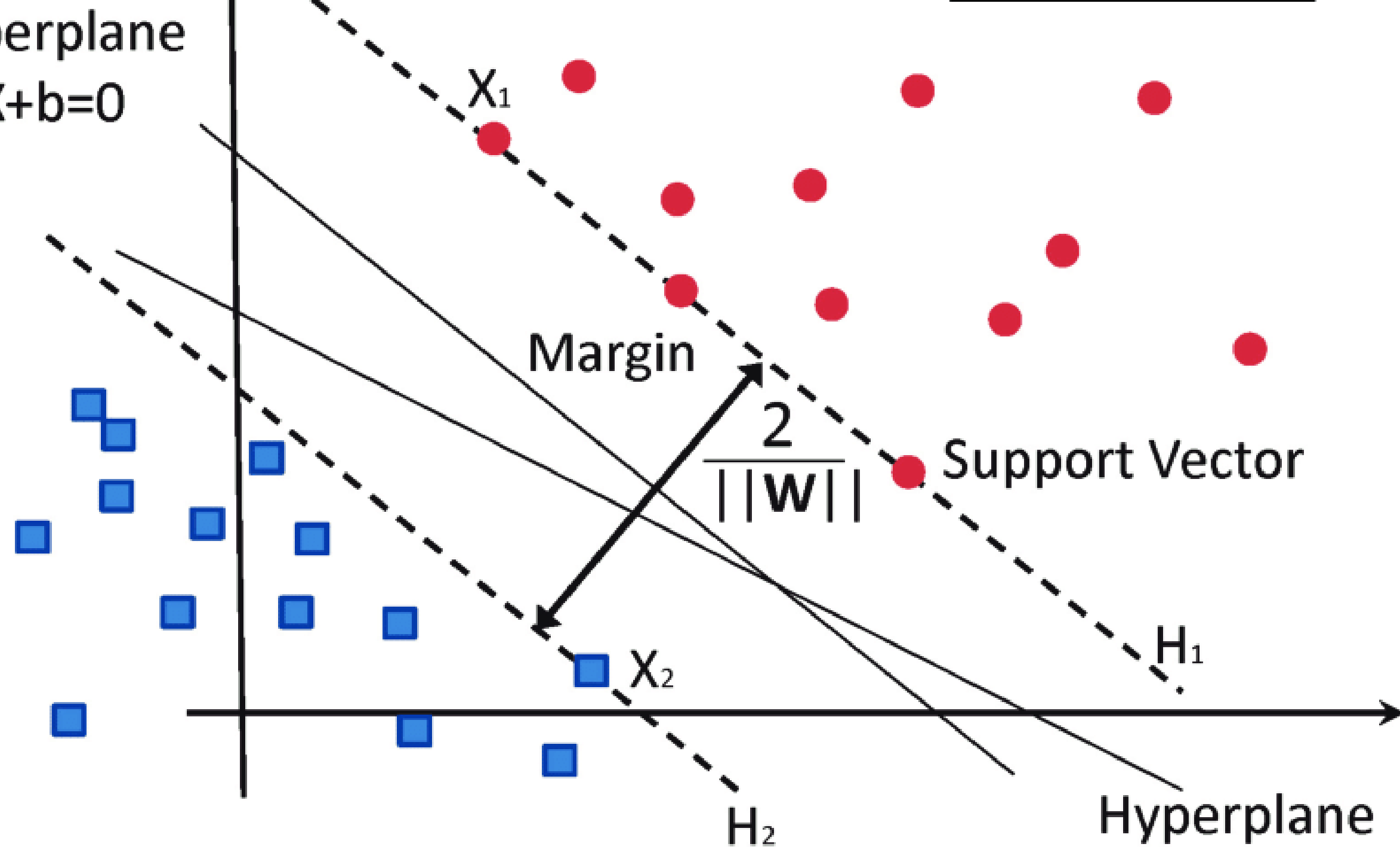




DISTINGUISHING CAT VS. DOG VIA COSINE SIMILARITY

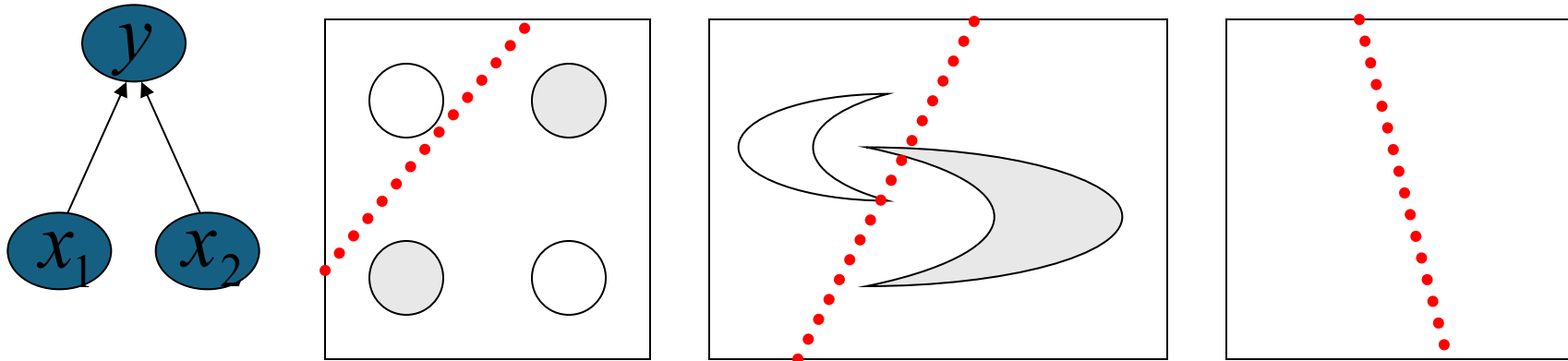


Hyperplane
 $W \cdot X + b = 0$



Decision Boundary

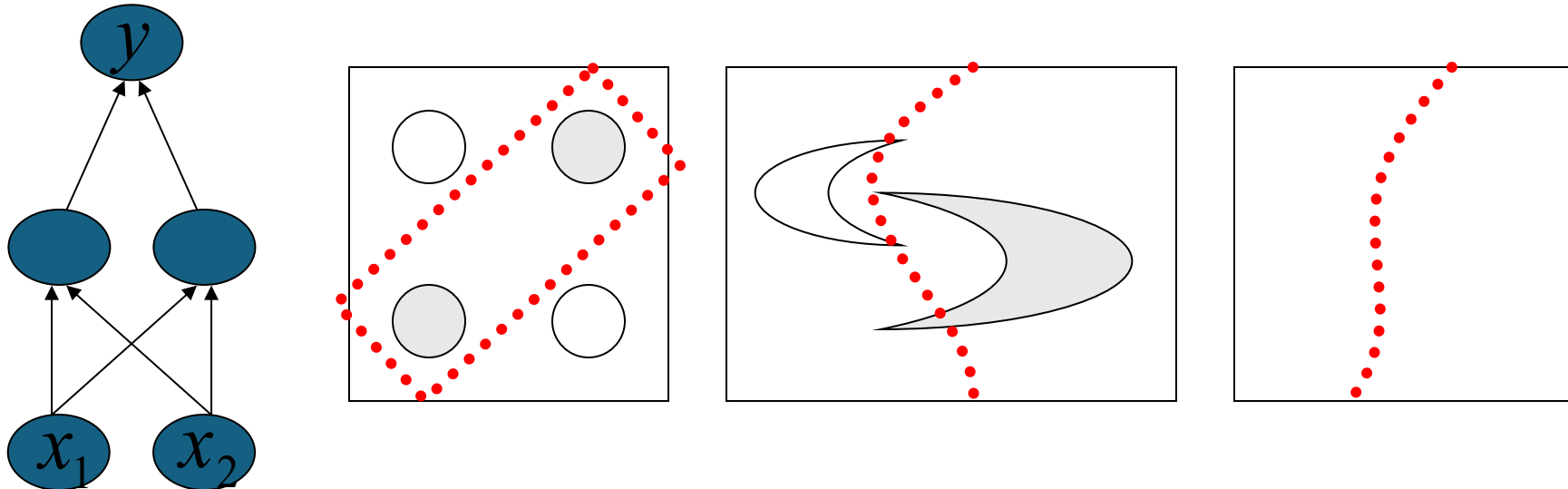
- 0 hidden layers: linear classifier
 - Hyperplanes



Example from to Eric Postma via Jason Eisner

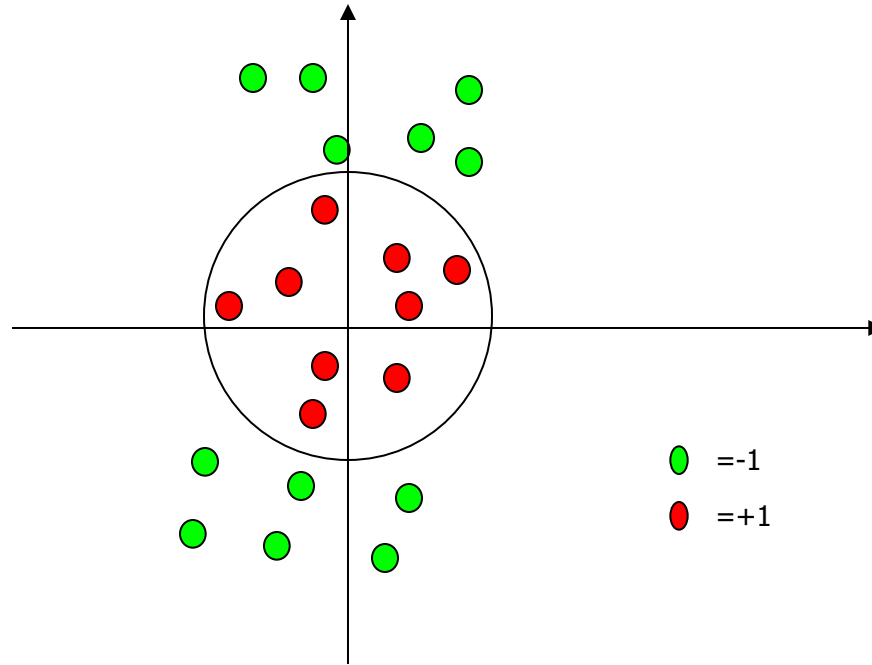
Decision Boundary

- 1 hidden layer
 - Boundary of convex region (open or closed)



Example from to Eric Postma via Jason Eisner

Problems with linear SVM



What if the decision function is not a linear?

M.C. Esher's Mathematical Art



M.C. Esher's Mathematical Art

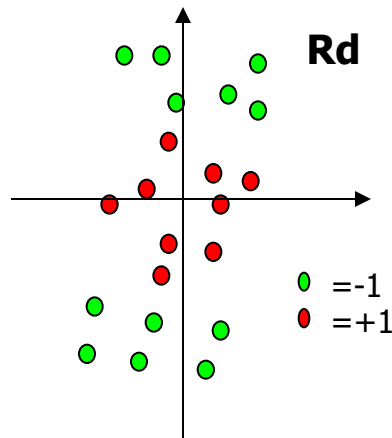


M.C. Escher's Mathematical Art



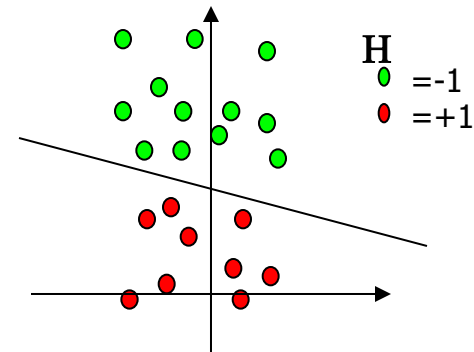
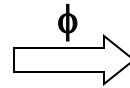
Non-linear SVM 1

The Kernel trick



Imagine a function ϕ that maps the data into another space:

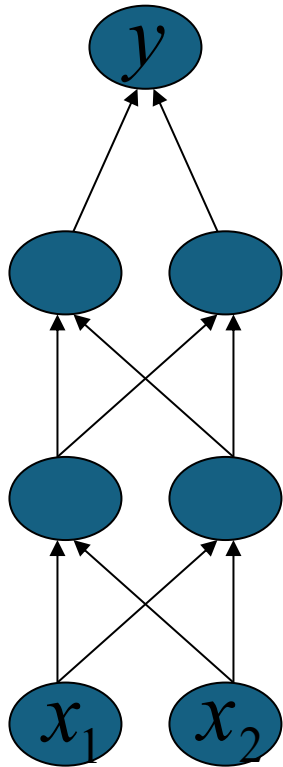
$$\phi: \mathbf{R}^d \rightarrow \mathbf{H}$$



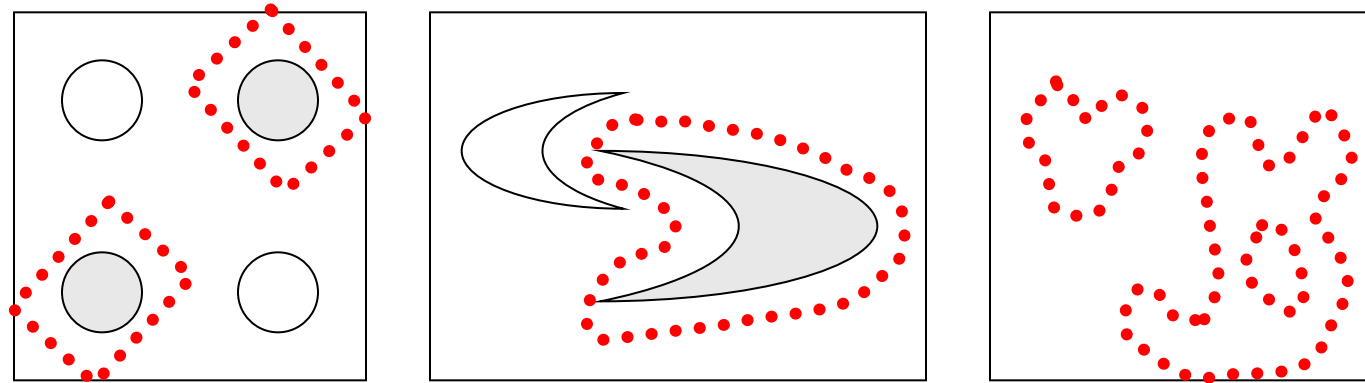
Remember the function we want to optimize: $L_d = \sum \alpha_i - \frac{1}{2} \sum \alpha_i \alpha_j x_i \cdot x_j$,
 x_i and x_j as a dot product. We will have $\phi(x_i) \cdot \phi(x_j)$ in the non-linear case.
If there is a "kernel function" K such as $K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$, we do not need to know ϕ explicitly. One example:

$$K(x, x') = \exp \left(- \frac{\|x - x'\|^2}{2\sigma^2} \right)$$

Decision Boundary



- 2 hidden layers
 - Combinations of convex regions



Vectors and Matrices: Basic Operation



- In a bold shift from tech's usual advice, Elon Musk and Nvidia CEO Jensen Huang are urging students to focus less on coding and more on physics and mathematics. In the age of artificial intelligence, they argue that deep understanding of the world's core principles will matter more than ever.

Robotics

English 

Announcing Newton, an Open-Source Physics Engine for Robotics Simulation

Mar 18, 2025

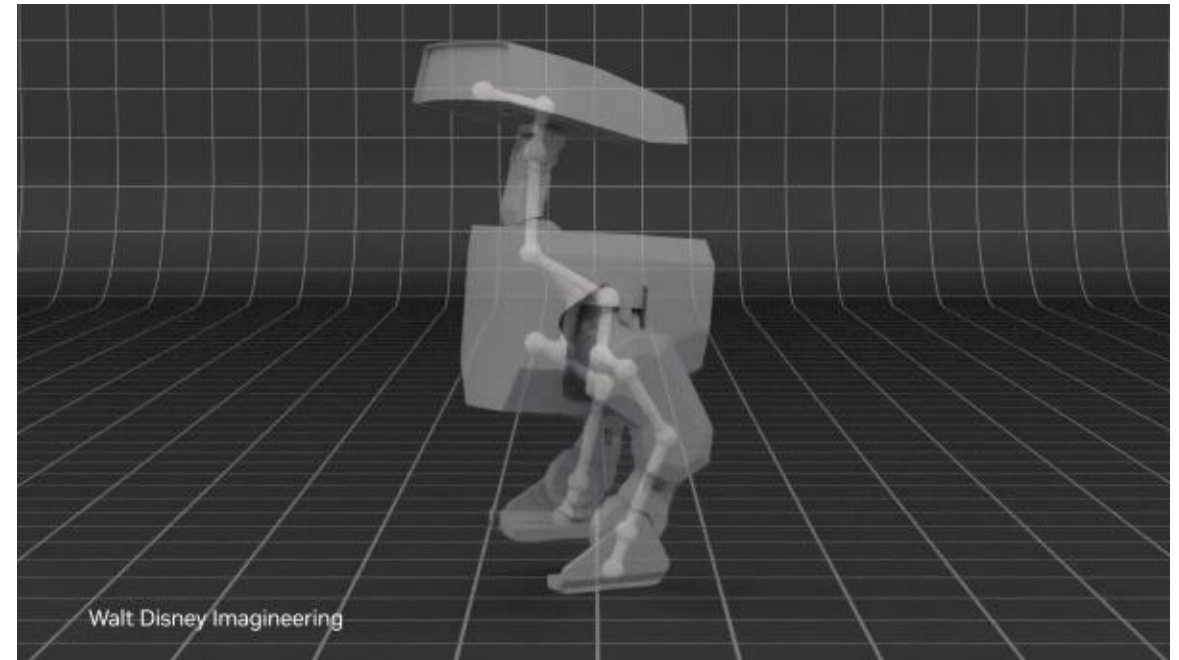
 +90 Like  Discuss (5)

By [Spencer Huang](#), [Miles Macklin](#) and [Gavriel State](#)

Physical AI models enable robots to autonomously perceive, interpret, reason, and interact with the real world. Accelerated computing and simulations are key to developing the next generation of robotics.

Physics plays a crucial role in robotic simulation, providing the foundation for accurate virtual representations of robot behavior and interactions within realistic environments. With these simulators, researchers and engineers can train, develop, test, and validate robotic control algorithms and prototype designs in a safe, accelerated, and cost-effective manner.

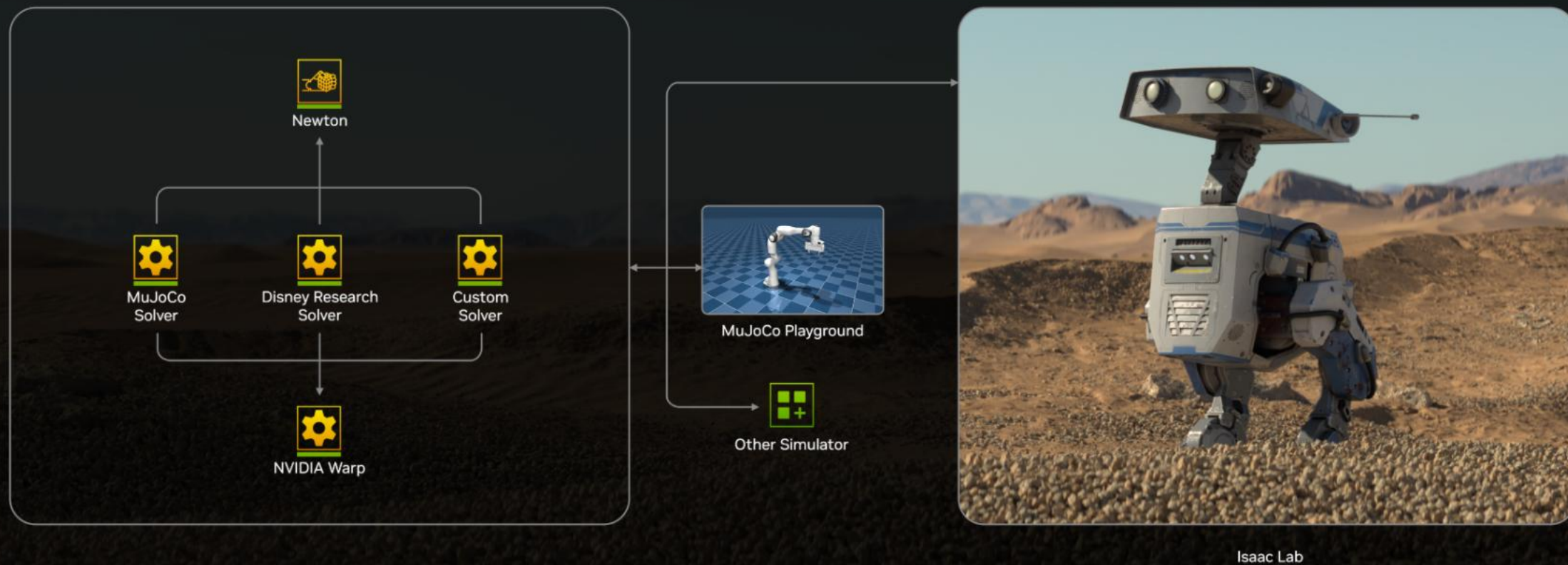
Robotic simulation is built upon the fundamental laws of physics, including the conservation of mass and momentum, rigid and soft body dynamics, contact and friction, and actuator modeling. These principles are used to predict how robots will behave in various scenarios and environments.



Disney Research

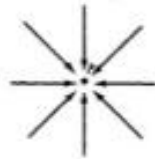
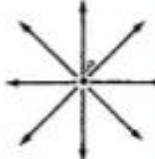
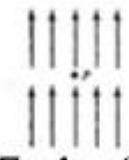
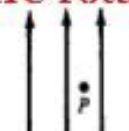
 NVIDIA

 Google DeepMind



Physical meaning

GRAD, DIV & CURL

	Gradient	Divergence	Curl
Physical meaning	A vector that gives direction of the maximum rate of change of a quantity i.e. temp	A scalar that measures the magnitude of a source or sink at a given point  $\nabla \cdot \mathbf{A} < 0$ Sink i.e. Flux out < flux in  $\nabla \cdot \mathbf{A} < 0$ Source i.e. Flux out > flux in  $\nabla \cdot \mathbf{A} = 0$ Incompressible Flux out = flux in	A vector operator that describes the rotation/ununiformity of a vector field  $\nabla \times \mathbf{A} > 0$ RHC rotation  $\nabla \times \mathbf{A} < 0$ LHC rotation  $\nabla \times \mathbf{A} = 0$ Irrotational

Gradient

Physical meaning

What is the Gradient of a Scalar?

The gradient is analogous to a derivative.

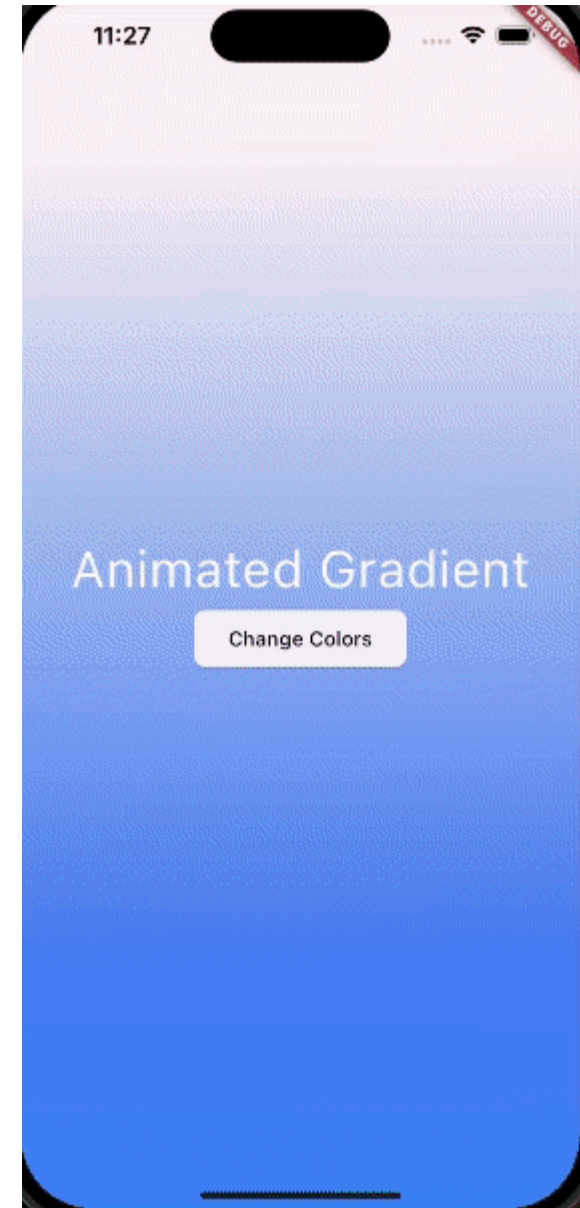
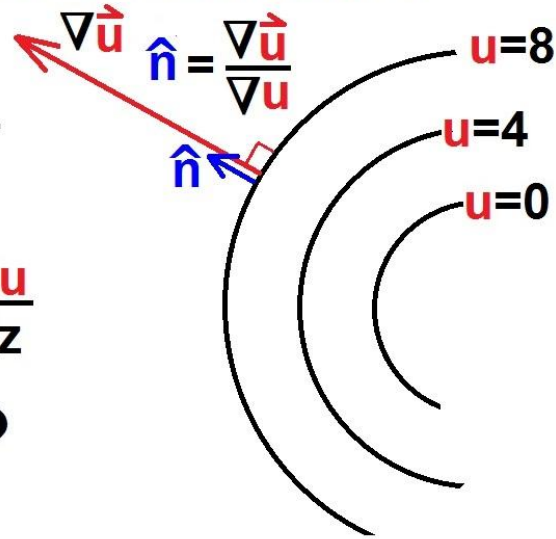
Del operator:

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

Gradient of u :

$$\nabla u = \hat{x} \frac{\partial u}{\partial x} + \hat{y} \frac{\partial u}{\partial y} + \hat{z} \frac{\partial u}{\partial z}$$

Ex: $u = y^2 - x$ $\nabla u = ?$



A Recipe for Machine Learning

1. Given training data:

$$\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^N$$

2. Choose each of these:

– Decision function

$$\hat{\mathbf{y}} = f_{\boldsymbol{\theta}}(\mathbf{x}_i)$$

– Loss function

$$\ell(\hat{\mathbf{y}}, \mathbf{y}_i) \in \mathbb{R}$$

3. Define goal:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^N \ell(f_{\boldsymbol{\theta}}(\mathbf{x}_i), \mathbf{y}_i)$$

4. Train with SGD:

(take small steps
opposite the gradient)

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta_t \nabla \ell(f_{\boldsymbol{\theta}}(\mathbf{x}_i), \mathbf{y}_i)$$

Key points:

1. Explore a **new class of decision functions** (Neural Networks)
2. Consider **variants of this recipe** for training

– Decision function

$$\hat{y} = f_{\theta}(x_i)$$

– Loss function

$$\ell(\hat{y}, y_i) \in \mathbb{R}$$

4. Train with SGD:

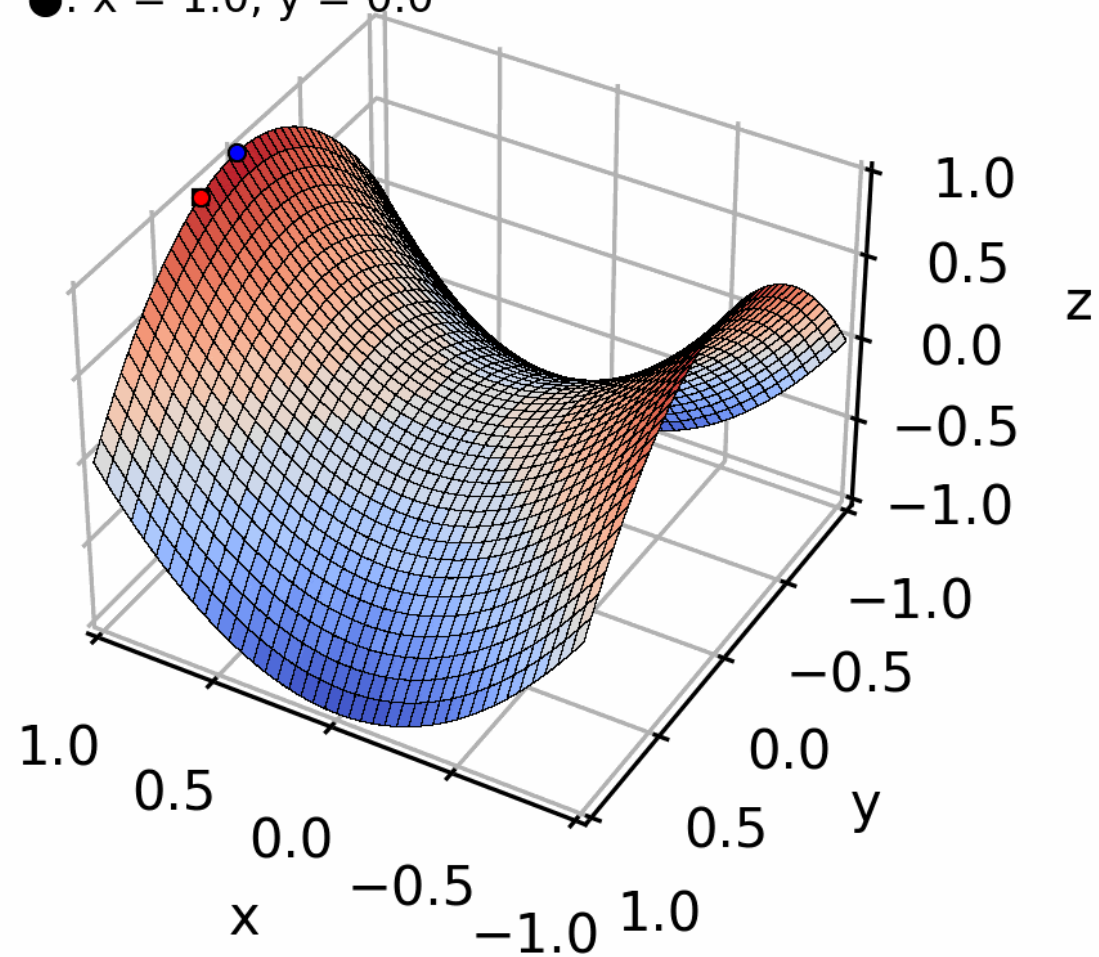
– Take small steps
opposite the gradient)

$$\theta^{(t+1)} = \theta^{(t)} - \eta_t \nabla \ell(f_{\theta}(x_i), y_i)$$

Initilization

■: $x = 1.0, y = 0.25$

●: $x = 1.0, y = 0.0$



Divergence

Example of Applied Divergence: Anti-static wrist strap



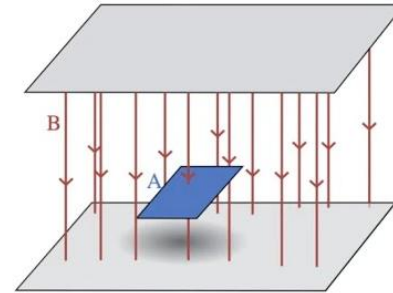
Gauss's Law for the Electric Field

differential form

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

integral form

$$\int \vec{E} \cdot \hat{n} \, da = \frac{q_{\text{enc}}}{\epsilon_0}$$



the **flux** that passes
through a closed surface
is proportional to the
charge enclosed therein

Curl

Physical Meaning of curl (rot)

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \omega_x & \omega_y & \omega_z \\ x & y & z \end{vmatrix} = (\omega_y z - \omega_z y)\mathbf{i} + (\omega_z x - \omega_x z)\mathbf{j} + (\omega_x y - \omega_y x)\mathbf{k}$$

$$v_x = \omega_y z - \omega_z y \quad \omega_x = \frac{1}{2} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right), \quad \omega_y = \frac{1}{2} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right), \quad \omega_z = \frac{1}{2} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

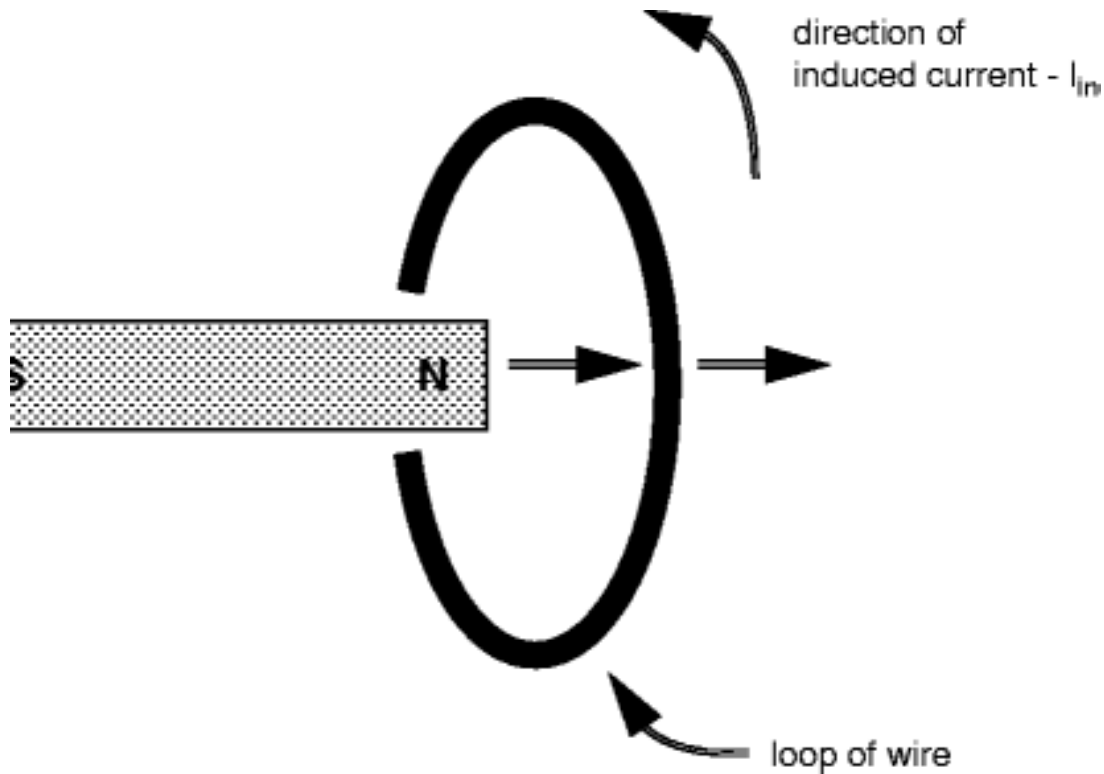
$$\begin{aligned} v_y &= \omega_z x - \omega_x z \\ v_z &= \omega_x y - \omega_y x \end{aligned} \quad \boldsymbol{\omega} = \frac{1}{2} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \frac{1}{2} \nabla \times \mathbf{v}$$

$$\text{curl } \mathbf{v} \equiv \text{rot } \mathbf{v} \equiv \nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \mathbf{i} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \mathbf{j} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \mathbf{k}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = 2\boldsymbol{\omega}$$

The curl of a velocity field is angular velocity field (x2).

P: Physical meaning



$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

Reminder that the del operator is a vector

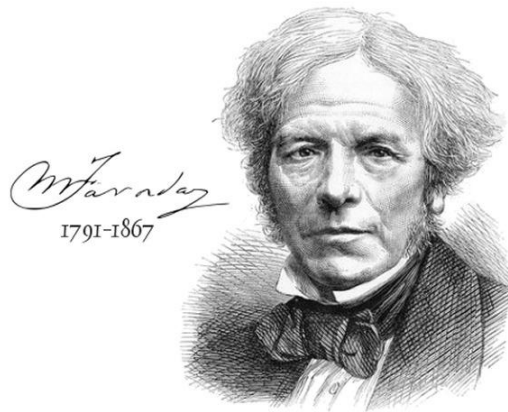
Reminder that the electric field is a vector

The differential operator called "del" or "nabla"

The electric field in V/m

The cross-product turns the del operator into the curl

The rate of change of the magnetic field with time



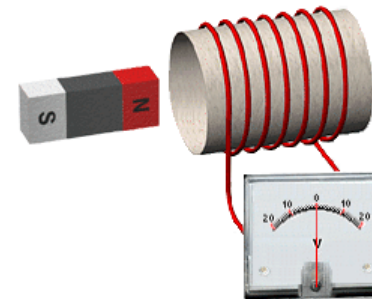
Integral form

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

Differential form

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Faradays Law of Induction



Kieran Mckenzie

P: Physical meaning

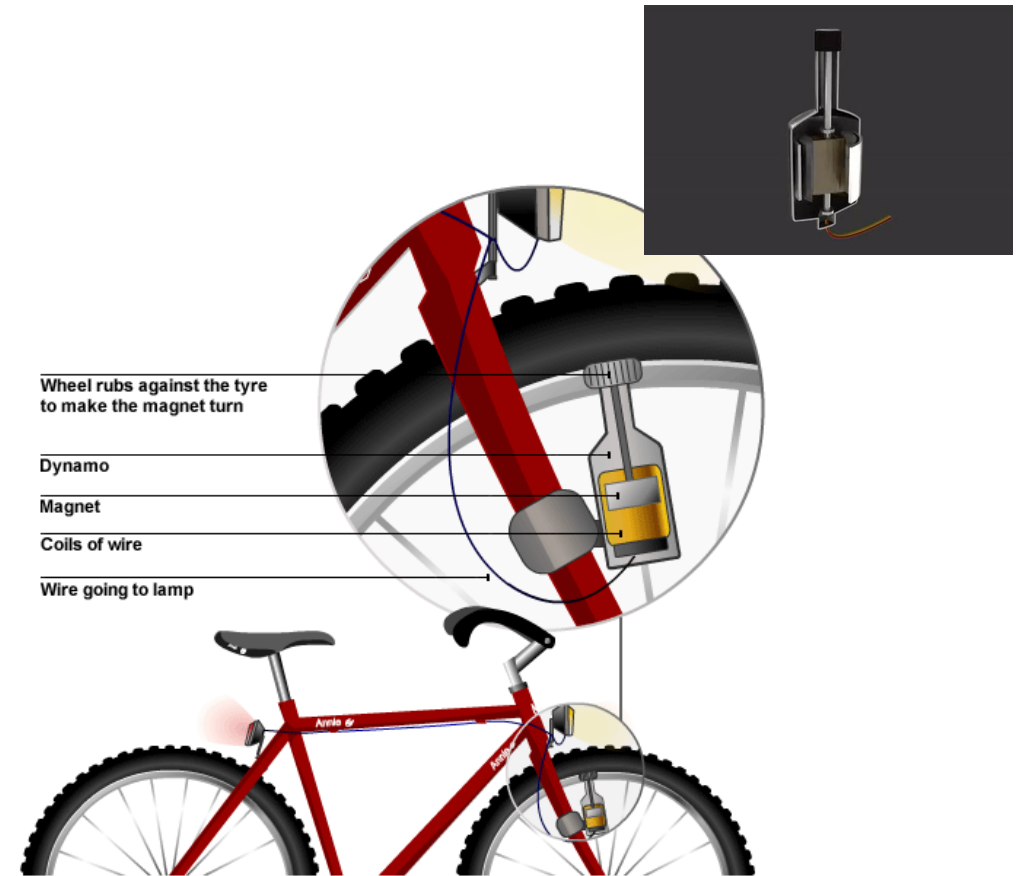


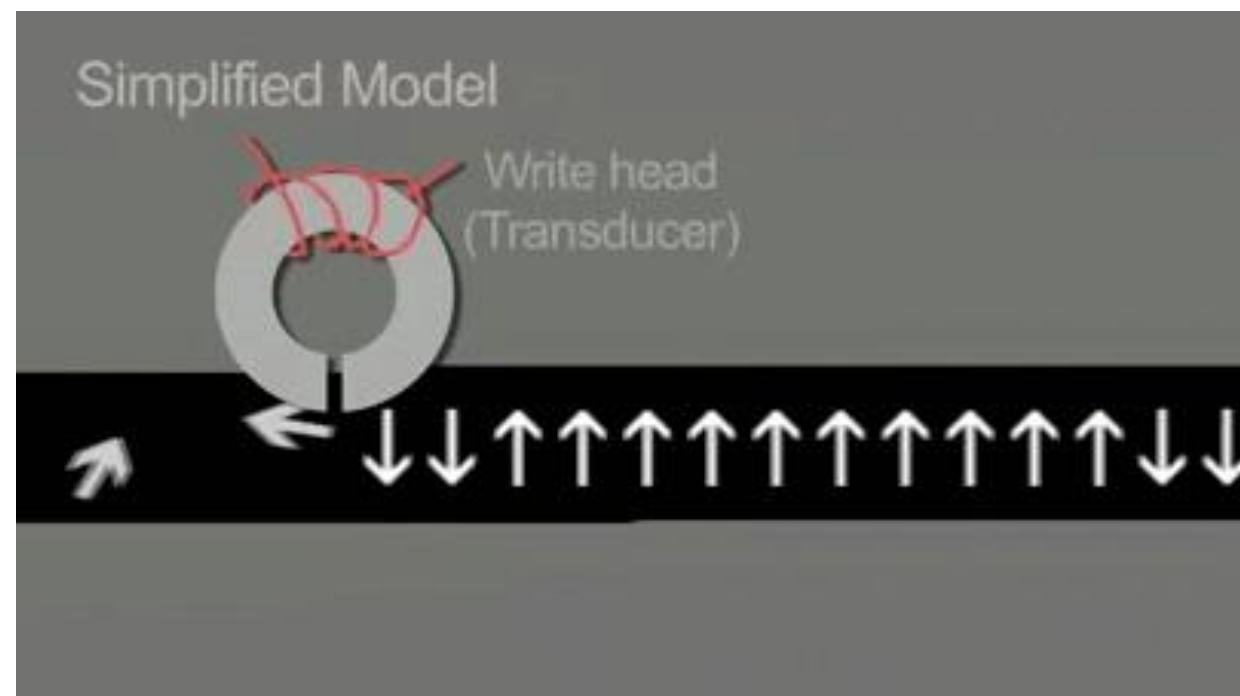
Induced electric field vector

$$\underbrace{\nabla \times \vec{E}}_{\text{Del cross operator means to take the curl}} = - \frac{\partial \vec{B}}{\partial t}$$

Del cross operator
means to
take the curl

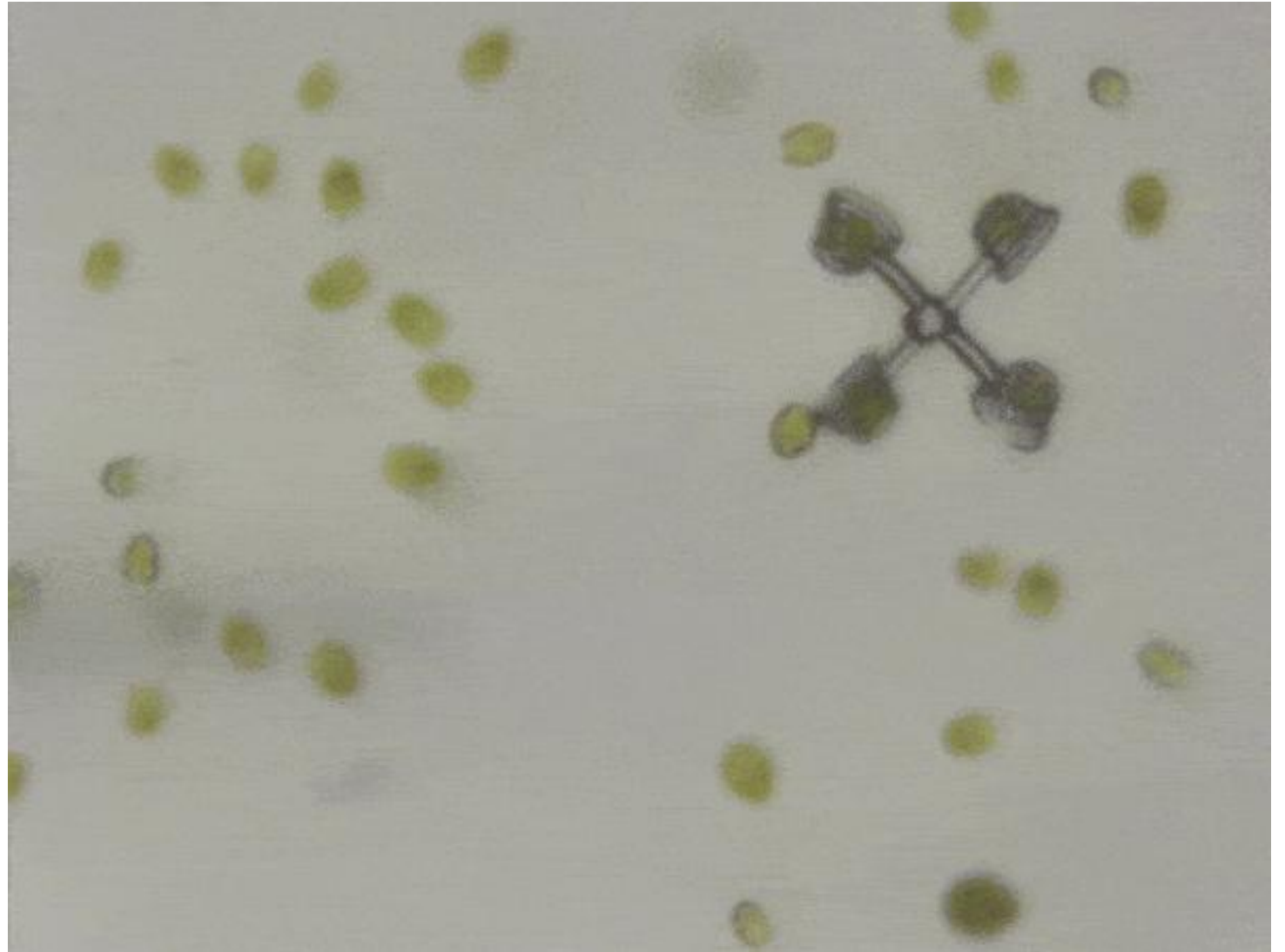
The rate of change
of the magnetic
flux density vector





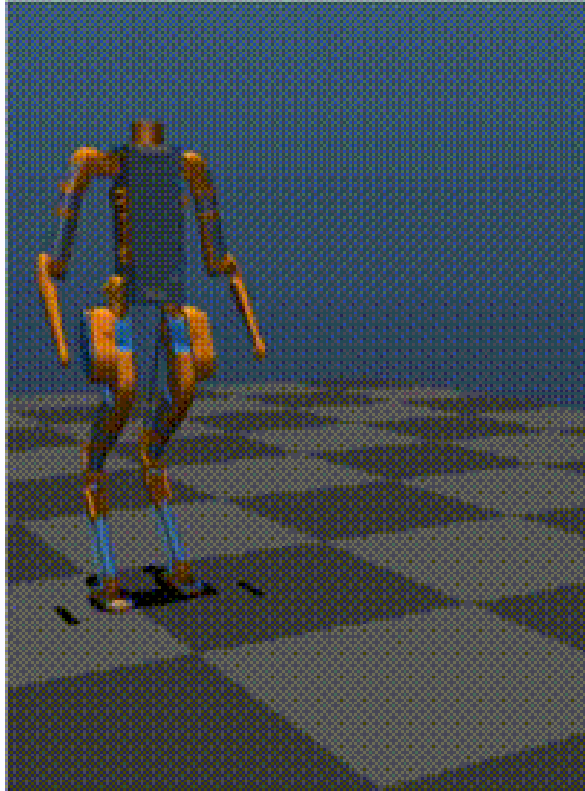
CURL:Example

- Haruka Oda, Naoto Shimizu, Yuya Morimoto, and Shoji Takeuchi, "Harnessing the Propulsive Force of Microalgae with Microtrap to Drive Micromachines," *Small*, 20, 2402923: July 7, 2024, doi:10.1002/sml.202402923.

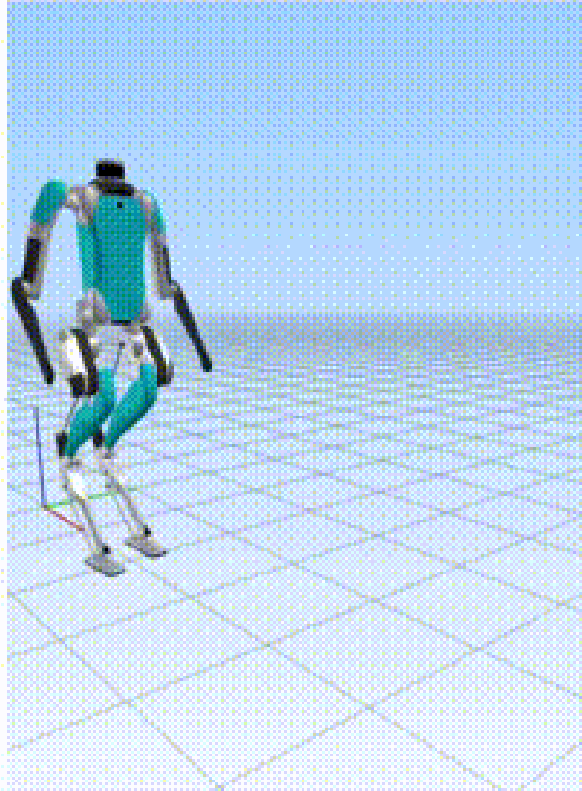


Microscopic rotator. Uninhibited, the algae can whizz along at over 100 micrometers per second. With four algae in the traps, the rotator moved at an average speed of between about 20-40 micrometers per second.
© 2024 The Shoji Takeuchi Research Group at the University of Tokyo

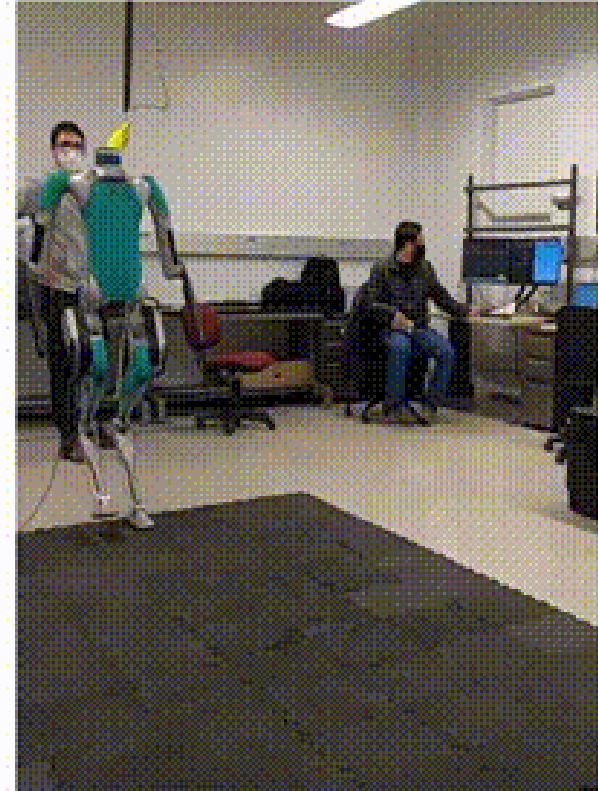
Custom simulation



AR simulation



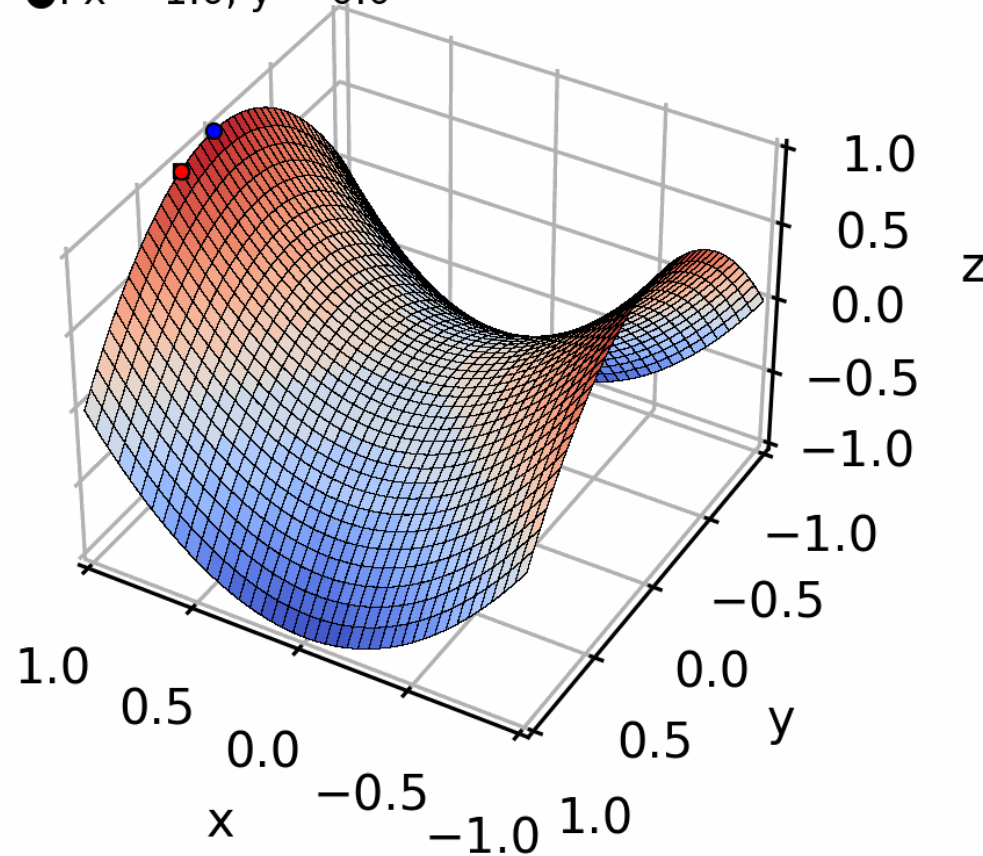
Hardware



Initialization

■: $x = 1.0, y = 0.25$

●: $x = 1.0, y = 0.0$



Gradient Descent Point

Fast Gradient Sign Method

Score of label y_{true} , given input image X

$$X^{adv} = X + \epsilon \operatorname{sign}(\nabla_X J(X, y_{true}))$$

Adversarial Machine Learning



x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

$=$

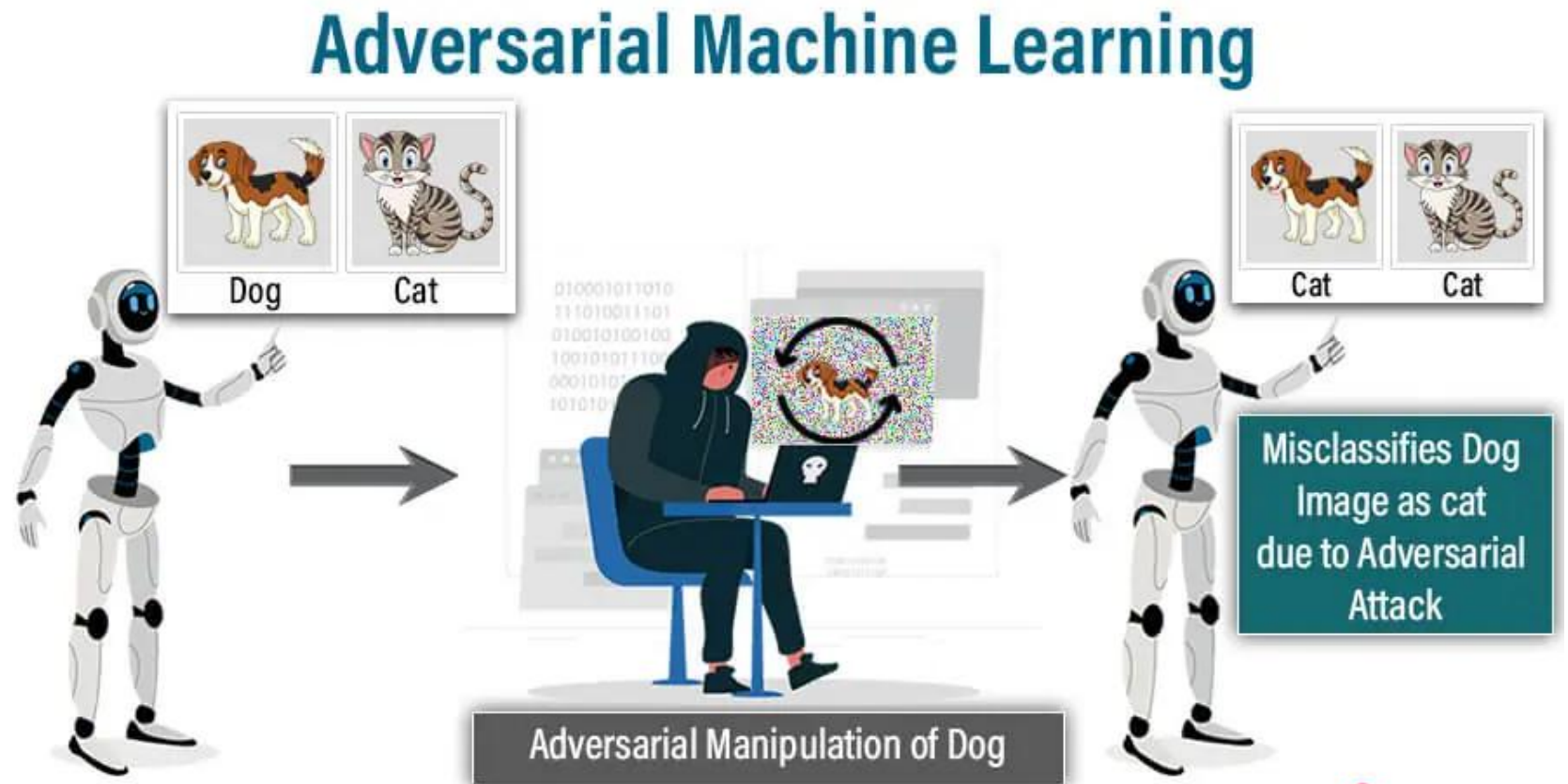


$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

99.3 % confidence

Adversarial Machine Learning





HIGH CONVENIENCE /
LOW SECURITY



CONVENIENCE $\propto 1$ / SECURITY

HIGH SECURITY /
LOW CONVENIENCE



To stay secure, we need algorithms to protect high-tech innovation!

To build algorithms, we need ideas and strong fundamentals of thinking!

That is why we need this course for the above reasons!



THANKS!

Any questions?

You can find me at
pitikhate@cmkl.ac.th